

How the Financing of Balance-Sheet Policy Shapes Its Effects*

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Abstract

I ask whether the financing of Federal Reserve balance-sheet policy shapes its effects on real GDP and the price level. In a monthly U.S. SVAR I identify two balance-sheet shocks, one financed by reserves and one by overnight reverse repurchases (ON RRP). Both balance-sheet contractions lower the price level, and only the reserve-financed contraction lowers GDP. The reserve-financed contraction alone raises the excess bond premium. Over the quantitative easing after the financial crisis and the first quantitative tightening the reserve-financed shock is the largest contributor to both forecast errors. It pushes both variables above their forecasts over the easing and pulls them below over the tightening. During the ON RRP drain of the second quantitative tightening the ON RRP-financed shock is the largest contributor to the price-level forecast error. It pulls the price level below its forecast. Which liability finances a balance-sheet policy shapes what that policy does.

Keywords: quantitative tightening, quantitative easing, central bank liabilities, overnight reverse repurchases, excess bond premium, zero restrictions, shock heteroskedasticity

JEL codes: C11, C32, E51, E52, E58, G21

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1 Introduction

A balance-sheet operation of the Federal Reserve moves the securities of the System Open Market Account (SOMA) on the asset side. The liability side moves by the same amount, in reserves, bank deposits at the Federal Reserve, or in overnight reverse repurchases (ON RRP), overnight loans from money market funds (MMFs). The Federal Reserve pays for a purchase with reserves at first. Those reserves stay in the banking system or move into ON RRP as the holders of the cash decide, within the terms the Federal Reserve sets for the facility (link 1 of Figure 1). I say that a claim finances the operation when it holds the balance at the end of that process and receives the interest the Federal Reserve pays on it. A purchase is reserve-financed or ON RRP-financed by where its balance came to rest, not by how the Federal Reserve paid.

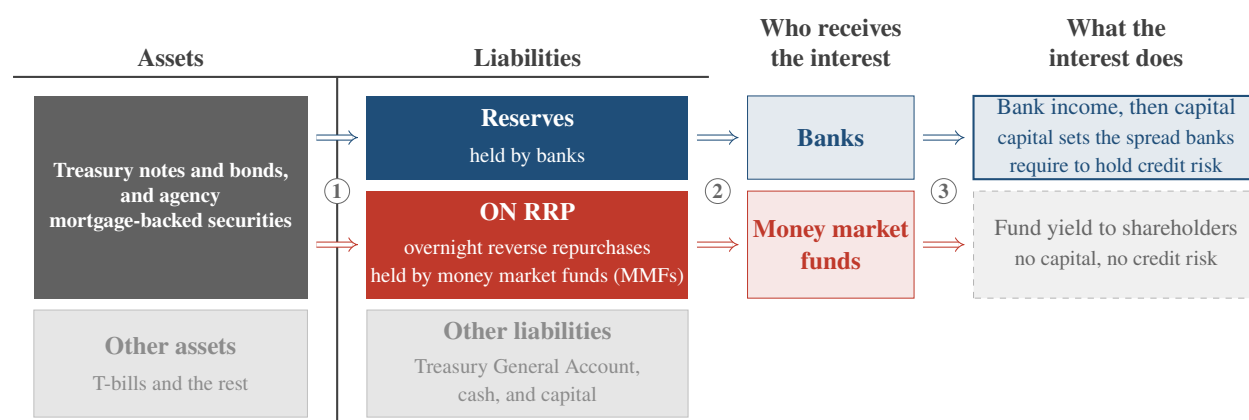


Figure 1: Two ways to finance the same operation, and two holders of the interest.

Note. A stylized Federal Reserve balance sheet. A purchase of securities on the asset side creates reserves, and the balance stays in reserves or moves into ON RRP as the holders of the cash decide. The third column names who receives the interest on each claim, and the fourth names what the interest does. The circled numbers mark the three links the text follows. Interest on reserves is bank income and adds to bank capital, which sets the spread banks require to hold credit risk. The ON RRP rate passes through money market funds to their shareholders as fund yield and builds no capital on the way. The reserve-financed chain ends at an intermediary whose capital limits the credit risk it carries, and the ON RRP-financed chain ends at its shareholders. Section 2 documents the two routes.

Link 2 of Figure 1 runs from the claim to the holder of the interest. The two ways of financing an operation send the Federal Reserve's interest to two different holders, interest on reserves to banks and the ON RRP rate to MMFs. Link 3 runs from the holder to what the interest does. For banks, the margin over what they pay on deposits is bank income, and that income adds to bank capital and to the credit risk banks can bear. MMFs pass the ON RRP rate to their shareholders as

fund yield, and a fund keeps none of it as capital and bears no credit risk. The same purchase feeds an intermediary that bears credit risk under one claim and no such intermediary under the other. This paper asks whether the transmission of a balance-sheet operation depends on which of the two claims finances the operation.

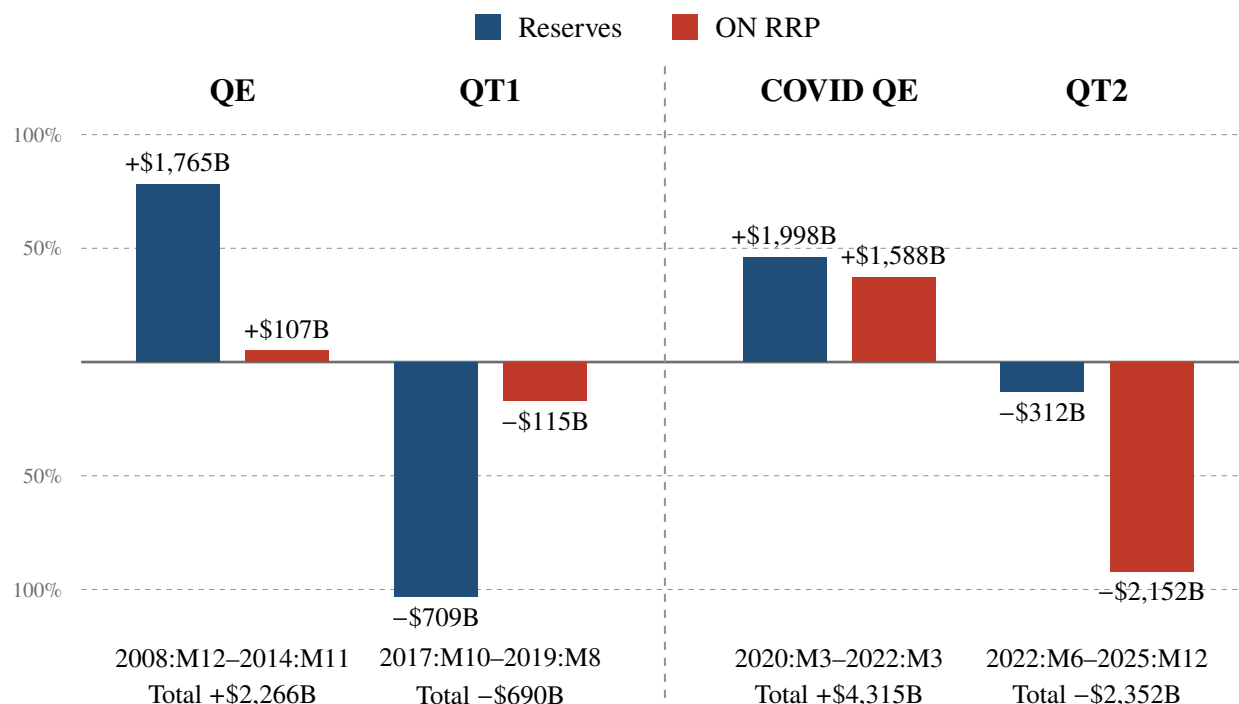


Figure 2: Liability transition across balance-sheet policy.

Note. Bars show the change in reserves and in ON RRP over the episode windows of Table A.1 as a percent of each episode’s total asset change, with QE episodes plotted upward and QT episodes downward. The number above each bar is the change in USD billions. The two shares do not sum to 100 percent, and the remainder is the change in the TGA, currency, and the residual liabilities, which is net positive in every episode. Reserves fell by more than total assets over QT1. Inside the COVID QE window the ON RRP buildup broadly matched the contemporaneous TGA drawdown, as Section 2 documents. Source: Federal Reserve H.4.1 and NY Fed SOMA holdings.

Since 2008 the Federal Reserve has run two rounds of quantitative easing (QE) and two of quantitative tightening (QT), and the claim that financed them changed from one round to the next. Reserves absorbed the 2008 to 2014 purchases, and QT1 drained them. Both claims absorbed the COVID purchases. QT2 left the reserves nearly intact and drained the ON RRP balances of MMFs instead (Figure 2). The four episodes are the same operation financed by the two claims of Figure 1. The transmission of balance-sheet policy to real GDP and the price level may differ with the claim. The question motivates two balance-sheet shocks, a reserve-financed shock (RFBS) and

an ON RRP-financed shock (OFBS).

I identify the two balance-sheet shocks in an eight-variable monthly U.S. SVAR, alongside demand, supply, and monetary policy (MP) shocks. The impulse responses give the first result. A balance-sheet contraction lowers the deflator whether reserves or ON RRP finance it, and only the reserve-financed contraction lowers GDP. The response of the excess bond premium (EBP), the spread intermediaries require to hold credit risk, says why. A reserve-financed operation moves the EBP, a contraction raising it and an expansion lowering it, and an ON RRP-financed operation does not move the EBP. The interest on reserves reaches an intermediary whose capital limits the credit risk it carries, and the spread that intermediary requires moves the cost of credit for firms. The ON RRP rate reaches no such intermediary, and the cost of credit does not move.

The forecast-error variance decomposition gives the share of the forecast variance that each shock accounts for at a given horizon. At three years the two balance-sheet shocks together account for 10% of the forecast variance of GDP and 11% of the forecast variance of the deflator. The EBP is the variable they account for most, at 22% together. The RFBS shock alone accounts for 18% of it, the largest share any labeled shock takes in any variable. The RFBS share of the GDP forecast variance is 7% against 3% for the OFBS shock.

The historical decomposition gives the contribution of each shock to the unexpected change in a variable over a window, and the windows are the volatility regimes of the model. Over the two regimes that reserves financed, 2009 to 2020, the RFBS shock is the largest contributor to the GDP, deflator, and EBP forecast errors. It pushes GDP and the deflator above their forecasts and the EBP below over the QE years and moves all three the other way over the years that hold QT1. Over the ON RRP drain of QT2 the OFBS shock is the largest contributor to the deflator forecast error and pulls the deflator below its forecast.

The identification needs four things. Two fit the sample, and for them I adopt the shock volatility and the heavy-tailed shocks of [Brunnermeier et al. \(2021\)](#). The two decades since 2008 contain the financial crisis, the COVID collapse, and the post-2021 inflation surge. The volatility of the shocks shifts across those phases, and the months around COVID are extreme within their phase.

Two label the shocks. Sign, zero, and narrative restrictions give the shocks their names. The zero restrictions carry the timing assumption of [Leeper et al. \(1996\)](#) and [Christiano et al. \(2005\)](#). Real activity and prices do not respond to a balance-sheet shock within the month. The balance sheet changes gradually after an announcement ([Nakashima et al., 2024](#)), and its real effects arrive after the quantities accumulate.

All four have to hold in one sampler, and no existing sampler imposes zero restrictions on a heteroskedastic model. This paper develops the heteroskedastic absorb-and-rotate sampler with zero restrictions (HARS-Z). HARS-Z extends the heteroskedastic absorb-and-rotate sampler (HARS) of [Kim and Zha \(2026\)](#), which combines the heteroskedastic likelihood of [Brunnermeier et al. \(2021\)](#) with sign restrictions and with the narrative restrictions of [Antolín-Díaz and Rubio-Ramírez \(2018\)](#).¹

To my knowledge, no existing paper estimates the real effects of balance-sheet policy separately by the liability that finances it. Most of the empirical literature measures the financial-market effects of balance-sheet policy in event studies around announcements, for QE ([Krishnamurthy and Vissing-Jorgensen, 2012](#); [Swanson, 2021](#)) and for QT ([Smith and Valcarcel, 2023](#); [Lloyd and Ostry, 2024](#); [Lu and Valcarcel, 2024](#)). Theoretical work models the transmission of balance-sheet policy ([Chen et al., 2012](#); [d’Avernas et al., 2024](#); [Kumhof and Salgado-Moreno, 2024](#); [Adrian et al., 2025](#)). In these models reserves, or the short-term debt of a consolidated government, finance the purchases. [Nakashima et al. \(2024\)](#) estimate the real effects of the Bank of Japan’s balance-sheet policy with a different identification. Their shocks separate the size of the balance sheet from its asset composition, not the liability side.

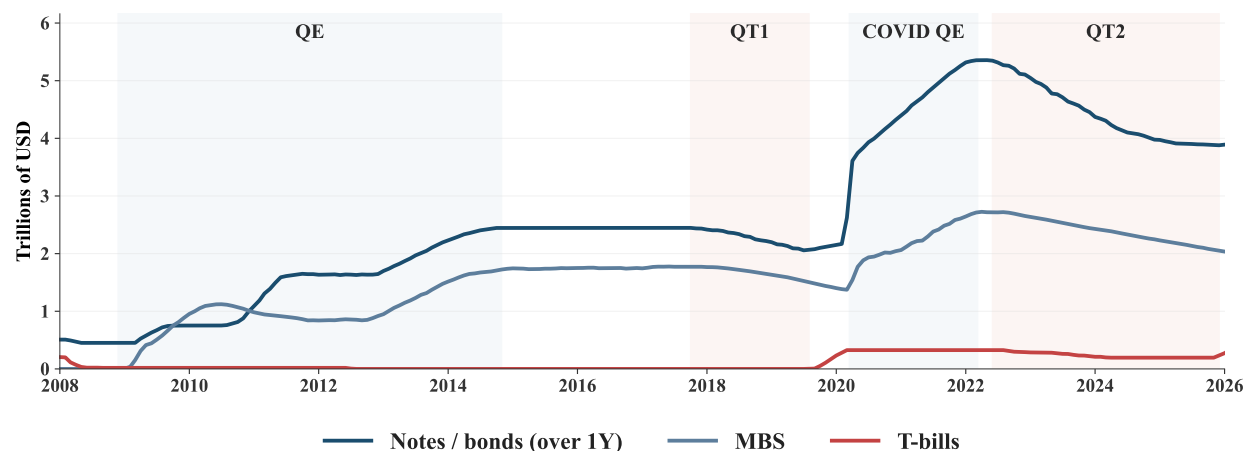
The rest of the paper proceeds as follows. [Section 2](#) documents the balance-sheet evidence. [Section 3](#) presents the empirical model, the data, and the identification. [Section 4](#) develops HARS-Z and establishes its validity. [Section 5](#) reports the impulse responses, their robustness, and the decompositions. [Section 6](#) concludes. The appendices contain the balance-sheet data, the estimation diagnostics, the formal statement of the narrative restrictions, the review of HARS and the theory of HARS-Z, the proofs, and the robustness exercises.

¹[Carriero et al. \(2024\)](#) combine the same heteroskedastic device with sign, narrative, and proxy restrictions. Neither sampler imposes zero restrictions.

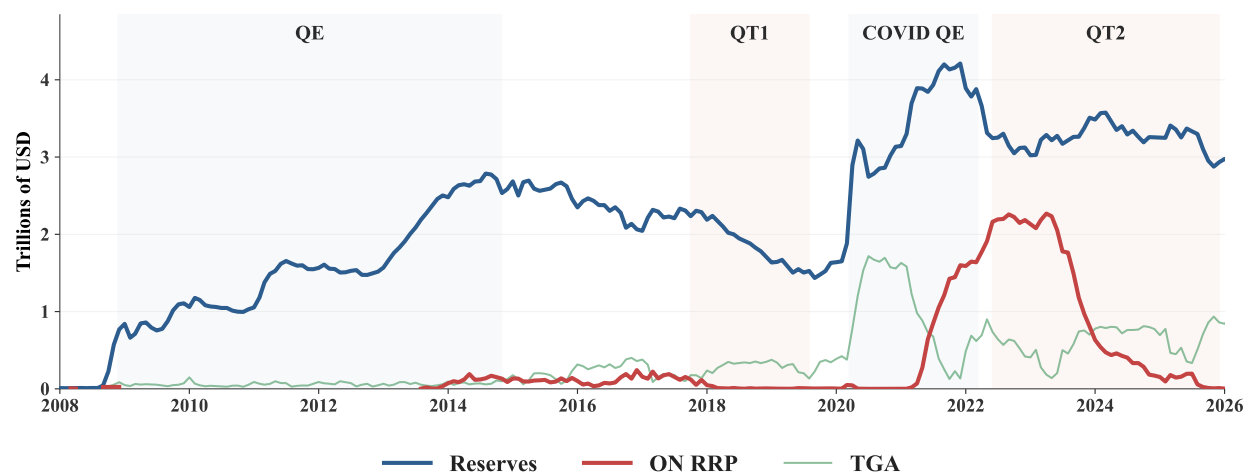
2 Two Ways of Financing the Balance Sheet

2.1 The Two Sides of the Balance Sheet

QE expands the Federal Reserve's balance sheet by buying securities, and QT shrinks it by letting securities run off at maturity without reinvestment. Figure 3 plots the assets and liabilities. Every QE raises both sides and every QT lowers them together. The balance-sheet identity fixes the totals and leaves open which liabilities absorb the change.



(a) Federal Reserve asset holdings.



(b) Federal Reserve liabilities.

Figure 3: The Federal Reserve's balance sheet.

Note. Panel (a) plots the Treasury and MBS holdings and panel (b) plots the reserve, ON RRP, TGA, and currency balances, in USD billions. The episode windows follow Table A.1. Source: Federal Reserve H.4.1 and NY Fed SOMA holdings.

The asset side. The four episodes differ in size and not in composition. Both expansions raised Treasury and MBS holdings together, and both contractions lowered them together (Figure 3a). The COVID QE raised total assets by \$4.3 trillion, roughly twice the \$2.3 trillion of the 2008 to 2014 expansion. QT2 ran longer and larger than QT1, with coupon runoff of \$1.47 trillion against \$0.38 trillion (Appendix A). The larger contraction followed the larger expansion.

The liability side. The episodes diverge in which liability they moved, and the process runs the same way in both directions. In a purchase the Federal Reserve pays the seller's bank in reserves, and the seller receives a deposit. The reserves stay where they are unless the deposit moves into a money market fund that places the cash in the ON RRP facility, which turns the reserves into ON RRP.² In a runoff the Treasury repays the maturing security from its account at the Federal Reserve and refills that account by issuing new debt. Reserves fall when banks or their depositors buy the new debt, and ON RRP falls when money market funds buy it with cash they take out of the facility. In both directions the holders of the cash decide which liability carries the balance, within the terms the Federal Reserve sets for the facility.

Each share below is percent of the episode's total asset change. Reserves absorbed 78% of the 2008 to 2014 expansion. QT1 reversed less than a third of that expansion and drained essentially the entire contraction from reserves. The COVID QE split between reserves (46%) and ON RRP (37%). QT2 drained ON RRP, 92% of the contraction, and left reserves nearly untouched at 13% (Appendix A).

2.2 The ON RRP Facility from Buildup to Drain

The ON RRP buildup. The ON RRP balance was \$154 billion when QT1 began and near \$2 trillion when QT2 began. The COVID purchases settled in reserves, and reserves kept rising into late

²Take a purchase of \$100 from a dealer. The dealer's bank receives \$100 of reserves, and the dealer receives a \$100 deposit. The dealer had borrowed from a money market fund in repo to hold the security, and it uses the deposit to repay the fund. The \$100 of reserves now sit at the fund's bank instead of the dealer's bank. If the fund places the \$100 at the ON RRP facility, the reserves leave the banking system and become ON RRP. If the fund buys a bill from another investor instead, the reserves stay in the banking system.

2021 as the buying continued (Figure 3b). The balance grew after March 2021, when the Treasury cut bill issuance and drew down the Treasury General Account (TGA) and the released cash flowed through money market funds (MMFs) into ON RRP (Wong, 2021). The expiry of the temporary Supplementary Leverage Ratio (SLR) relief in March 2021 added to it.³

The QT2 drain and the ON RRP facility. QT2 drained the ON RRP facility where QT1 had drained reserves. The Committee’s 2022 principles state that the runoff ends when reserves reach the level it judges consistent with an ample reserves regime (Federal Open Market Committee, 2022b). The drain ran from mid-2023, a year after the runoff began. Heavy Treasury bill issuance pushed T-bill yields above the ON RRP rate, and MMFs pulled their cash out of the facility and bought bills. Government funds left first. Their share of the balance fell from 80 percent in 2023:M7 to 35 percent in 2025:M9 (Figure A.1). The shift lowered ON RRP balances and spared reserves. Over the first three years of the runoff, securities holdings fell by \$1.3 trillion while reserves rose by \$350 billion (Logan, 2024).

Members of the Federal Open Market Committee watched the ON RRP facility as they set the pace of the runoff. Waller (2024) read the cash remaining in the facility as excess liquidity and as room to reduce securities holdings further. Logan (2024) attributed the decline in the facility to Treasury bill issuance and warned that further runoff would fall on reserves once the facility reached a low level. The Committee saw the same risk. In June 2024 it lowered the monthly Treasury redemption cap from \$60 billion to \$25 billion so that the runoff would not fall sharply on reserves (Federal Open Market Committee, 2024).

The ON RRP claim also moves duration. When reserves finance a purchase, the Federal Reserve takes duration out of private hands and long-term yields fall (Krishnamurthy and Vissing-Jorgensen,

³The SLR requires large banks to hold tier 1 capital against total leverage exposure without risk weights. In April 2020 the Federal Reserve temporarily excluded Treasuries and reserve balances from the exposure measure to ease strains in the Treasury market. The exclusion expired as scheduled on March 31, 2021, and reserves and Treasuries again counted toward bank leverage. Banks discouraged large deposits, the cash moved to MMFs, and the funds found few bills to buy (Afonso et al., 2025). Wong (2021) attributes the largest share of the ON RRP buildup to the fall in bill supply during the TGA drawdown, with the SLR expiry and the June 2021 rate adjustment adding to it.

2012; [Chen et al., 2012](#)). When it lets those securities run off, private investors take the duration back ([Crawley et al., 2022](#); [Wei, 2022](#)). A reader may take the ON RRP-financed episodes differently, with the buildup as MMFs parking cash and the drain as MMFs exchanging repo with the Federal Reserve for bills, and no duration moving either way.

Duration moved in both ON RRP episodes. The COVID purchases took coupons and MBS out of private hands whether the balance settled in reserves or moved into ON RRP. Over QT2 the Federal Reserve let \$1.5 trillion of Treasury coupons run off ([Table A.1](#)). Private investors hold those coupons instead, as they hold the coupons that ran off under QT1. [Table A.2](#) records where money market funds put the cash they took out of the facility, into Treasury securities and into repo lending to the hedge funds that hold Treasuries.⁴ Both financings move duration the same way, and [Section 3](#) imposes the same term-spread restriction on both balance-sheet shocks for that reason.

2.3 Why the Financing Should Matter

Each financing pays a different holder. The Federal Reserve pays interest on both claims, and the two financings differ in who receives it and in what that interest does. Only banks can hold reserves, and interest on reserves goes to them. A reserve-financed operation changes the quantity of reserves on bank balance sheets, and banks earn the rate on reserves on that quantity. Banks do not choose that quantity. The Federal Reserve sets it, and reserves cannot leave the banking system. What banks choose is how the rest of their balance sheets adjust, and [Diamond et al. \(2024\)](#) estimate that the adjustment is small. The reserves QE injected displaced a small fraction of bank lending to firms.⁵ The interest on that balance, net of its funding cost, is bank income. That income adds to bank capital, and capital limits how much credit risk, the risk that borrowers do not repay, a bank

⁴The figure and the tables of [Appendix A](#) are sector aggregates and show neither whom MMFs bought from nor the path of any dollar. MMFs may hold Treasury coupons with remaining maturity under thirteen months, and the data do not split Treasury holdings by class.

⁵[Diamond et al. \(2024\)](#) estimate a structural model of the deposit, mortgage, and loan markets on 2001 to 2017 data, with natural-disaster and Bartik instruments. In it, holding reserves carries a balance-sheet cost. In their counterfactual for the reserves QE injected from 2008 to 2017, loan rates rise and bank lending to firms falls by a small fraction of the reserves injected. Deposit rates rise and deposit quantities change little. The spread between the rate on reserves and the risk-free rate open to non-banks rises with the quantity of reserves in their model and in the data.

can carry.

The ON RRP rate goes to whoever holds the balance in the facility, and the holders are money market funds, most of them government funds.⁶ A fund keeps none of that interest as capital and passes it to its shareholders as fund yield. Nor does the operation change that yield. The fund earns the higher of the award rate and the bill yield wherever the balance sits. Moving the balance between the facility and bills changes what the fund holds, not what it earns. The yield reaches households and firms as interest income, of the same kind that depositors receive under reserve financing, and it builds no capital on the way.⁷ Nothing in the ON RRP-financed chain changes how much credit risk anyone can carry.

What separates the two financings is what the operation changes. A reserve-financed operation changes the quantity of an interest-bearing asset that banks hold, and the bank's income moves with it. An ON RRP-financed operation changes the composition of fund portfolios and not their size, and the fund's yield does not move. The bank keeps a margin between two rates, and the fund keeps none. The reserve-financed chain ends at an intermediary whose capital limits the credit risk it carries, and the ON RRP-financed chain ends at its shareholders. If the two chains end in different places, the difference should appear first in the cost of credit. Intermediaries with more capital lend to the same borrower at a smaller spread, and intermediaries with less capital at a larger one.⁸ Section 5 asks whether the difference shows in that spread and in GDP and the price level.

The terms and the realized split. The Federal Reserve set the terms that govern the split between reserves and ON RRP. It set the operating framework, the ON RRP award rate, the per-counterparty

⁶At the December 2022 peak MMFs held 92 percent of the ON RRP balance, and government funds, whose assets are Treasuries, agency debt, and repo, held 77 percent. Prime funds, which also buy commercial paper and certificates of deposit, never held more than 17 percent (Figure A.1).

⁷In the Financial Accounts, households and nonprofit organizations held 57 percent of money market fund shares at the end of 2022, and nonfinancial businesses held 17 percent (table F.521). The rest is spread over pension funds, other funds, insurers, and foreign holders. The table pools fund types, and government funds were 78 percent of fund assets on the same date (table S.123) ([Board of Governors of the Federal Reserve System, 2026](#)).

⁸Bank capital reaches that spread by two routes. The dealers who make markets in corporate bonds are for the most part affiliates of banking groups, twenty-one of the twenty-six primary dealers on the New York Fed's list ([Federal Reserve Bank of New York, 2026](#)). The inventory they can carry is bank capital. Banks are also the largest lenders to firms, and firms move from loans to bonds when banks tighten their terms ([Becker and Ivashina, 2014](#)). A bank with less capital pushes borrowers into the bond market and raises the spread there.

caps, and the runoff caps ([Federal Open Market Committee, 2022b,a](#)). Within those terms, Treasury issuance, bank regulation, and the portfolio choices of MMFs affected the realized split, as [Section 2.2](#) records ([Afonso et al., 2025](#)). Those terms set the financing. If the transmission differs with the financing, they also shape what an operation of a given size does to GDP and the price level. This paper does not model that choice. I take the realized financing as given and identify two balance-sheet shocks, one financed by reserves and one financed by ON RRP.

The gap in existing work. Existing work prices balance-sheet policy without separating the liability that finances it. [Crawley et al. \(2022\)](#) in an FRB/US simulation and [Wei \(2022\)](#) in a preferred-habitat model convert a runoff into an equivalent policy-rate increase. Both price it through the duration risk that private investors absorb. [Wei \(2022\)](#) traces which liability a runoff drains, reserves when banks buy the reissued coupon securities and ON RRP when MMFs buy bills. He does not estimate the two cases. In [Kumhof and Salgado-Moreno \(2024\)](#) and [d’Avernas et al. \(2024\)](#) reserves finance balance-sheet policy and banks transmit it, and neither model includes an ON RRP facility.

3 Empirical Model

Let y_t be the $n \times 1$ vector of endogenous variables and x_t the $r \times 1$ vector of predetermined terms, including the constant and p lags of y_t . The structural equation is

$$y_t' A_0 = x_t' A_+ + \eta_t', \quad (1)$$

where A_0 is the $n \times n$ nonsingular structural impact matrix and A_+ is the $r \times n$ matrix stacking the constant and lag coefficients. The vector η_t collects the model shocks, independent across equations and time conditional on the regime.⁹

⁹Under this right-multiplication convention, which (1) shares with [Kim and Zha \(2026\)](#) and [Arias et al. \(2018\)](#), columns of A_0 correspond to equations.

Following the deterministic variance-regime specification of [Brunnermeier et al. \(2021\)](#), let

$$s_t : \{1, \dots, T\} \rightarrow \{1, \dots, h\}$$

be a known date-to-regime map that assigns each date to one of h volatility regimes. The shocks are heteroskedastic across regimes and heavy-tailed within them,

$$\eta_{i,t} \mid \nu_{i,t}, s_t = k \sim \mathcal{N}(0, \lambda_{i,k} \nu_{i,t}), \quad i = 1, \dots, n, \quad (2)$$

where $\lambda_{i,k} > 0$ is the relative variance of shock i in regime k and $\nu_{i,t}$ is a latent scale with an inverse-gamma distribution, as in [Brunnermeier et al. \(2021\)](#). The inverse-gamma parameters make each standardized shock a Student- t with unit scale, and the variance of shock i in regime k is proportional to $\lambda_{i,k}$. I normalize the regime-specific variances to satisfy

$$\frac{1}{h} \sum_{k=1}^h \lambda_{i,k} = 1, \quad i = 1, \dots, n,$$

which fixes the scale of each shock variance path across regimes.

Post-multiplying (1) by A_0^{-1} gives the reduced form

$$y'_t = x'_t B + u'_t, \quad B = A_+ A_0^{-1}, \quad u'_t = \eta'_t A_0^{-1}, \quad (3)$$

where the regime-specific reduced-form covariances

$$\Sigma_k = \text{Var}(u_t \mid s_t = k) = (A_0^{-1})' \Lambda_k A_0^{-1}, \quad \Lambda_k := \text{diag}(\lambda_{1,k}, \dots, \lambda_{n,k}), \quad (4)$$

identify the columns of A_0 up to signed permutation whenever the variance ratios across regimes are not mutually proportional ([Rigobon, 2003](#); [Lanne et al., 2010](#)). The covariances identify (A_0, Λ) but do not label the shocks. In this paper, the economic labeling comes from the restricted rotation, and the two roles stay separate throughout.

3.1 Data and Priors

I estimate the model of [Section 4](#) on a monthly U.S. system with eight variables ($n = 8$) and $p = 6$ lags. The sample is November 2008 through March 2026.¹⁰

The system contains eight variables. The first four are the one-year Treasury yield, the log of SOMA holdings, $\log(\text{Reserves} + \text{ON RRP})$, and the ON RRP ratio $\text{ON RRP}/(\text{Reserves} + \text{ON RRP})$. SOMA holdings cover Treasury notes and bonds plus agency MBS and exclude bills.¹¹ The last four are the log of real GDP, the log of the GDP deflator (GDPDEF), the term spread, and the excess bond premium (EBP). The term spread is the 10-year Treasury yield less the 3-month bill rate. I map quarterly GDP and the deflator to monthly frequency by Chow–Lin interpolation and aggregate every series to monthly means.

The one-year Treasury yield stands in for the policy rate. At the zero lower bound the funds rate carries no innovation. [Gertler and Karadi \(2015\)](#) take a short-maturity government yield as the policy indicator, on the ground that announcements about future policy operate over a horizon of about two years. Their monthly baseline uses the one-year.

The EBP is in the system for the reason [Section 2](#) gives, which turns on the spread intermediaries require to hold credit risk. The EBP measures the financial sector’s capacity to bear that risk ([Gilchrist and Zakrajšek, 2012](#)).¹² [Glover and Oliyide \(2024\)](#) read it as the willingness of lenders to lend beyond what the riskiness of the loans explains. [Gertler and Karadi \(2015\)](#) include it in a system like this one.

[Section 2](#) implies a prediction for the EBP. Suppose the reserve-financed chain adds to the income and capital of intermediaries and the ON RRP-financed chain does not. Intermediaries with

¹⁰The November 2008 start coincides with the financial-crisis interventions that preceded QE1. It drops the pre-crisis regime, where reserves stayed near zero and reserve variance was orders of magnitude smaller.

¹¹The Federal Reserve treats bills separately from the programs. It purchased bills from October 2019 into 2020 to keep reserves ample. It described those purchases as technical measures that do not represent a change in the stance of monetary policy ([Federal Open Market Committee, 2019](#)). Under the 2022 runoff the decline in Treasury holdings covers coupon securities, and bills fall only to the extent that coupon maturities run below the monthly cap ([Federal Open Market Committee, 2022a](#)). Bill holdings track the maturity schedule and the reserve-management purchases rather than the purchase and runoff decisions.

¹²[Gilchrist and Zakrajšek \(2012\)](#) take the corporate bond spread and remove the part that the measured default risk of the issuers accounts for. What they remove moves with the borrowers, and what remains moves with the lenders. They read a rise in the remainder as a fall in the financial sector’s capacity to bear risk.

more capital lend at a smaller spread, and a reserve-financed expansion would lower the EBP while a reserve-financed contraction would raise it. An ON RRP-financed operation would move the EBP in neither direction. The EBP enters as an outcome and carries no restriction, and the data decide whether that pattern appears. [Section 5.1](#) reports how each balance-sheet shock moves it.

Two liability variables enter the system. The size variable $\log(\text{Reserves} + \text{ON RRP})$ sums the two liabilities that absorbed the QE and QT episodes of [Section 2](#). It treats reserves and ON RRP as close substitutes, two forms of overnight central-bank money that commercial banks and money market funds hold, respectively ([Diamond et al., 2024](#); [Copeland et al., 2021](#)). The ON RRP ratio $\text{ON RRP}/(\text{Reserves} + \text{ON RRP})$ measures the sector split of that total. The ratio distinguishes the two contractions. A reserve drain and an ON RRP drain reduce the size variable by the same amount, but they move the ratio in opposite directions. The tables abbreviate the two variables as $\log(R + \text{RRP})$ and $\text{ON RRP}/(R + \text{RRP})$.¹³

The priors follow [Sims and Zha \(1998\)](#) and [Brunnermeier et al. \(2021\)](#). Each element of A_0 has an independent Gaussian prior with mean 100 on the diagonal, 0 off the diagonal, and standard deviation 200. The centering at $A_0 = 100I$ scales each equation to an expected prior standard deviation of 0.01, and the standard deviation of 200 keeps the prior diffuse. A Dirichlet($\alpha = 2$) prior on the normalized variance shares centers the prior on equal variances across regimes. It enforces the normalization that the relative variances of each shock average to one.

Sims–Zha dummy observations center each equation on an independent random walk, except the ON RRP ratio and the term spread, which I center on white noise.¹⁴ Following [Brunnermeier et al. \(2021\)](#), the latent scales are i.i.d. inverse-gamma and make each standardized shock Student- t with 4.99 degrees of freedom. [Section 3.2](#) reports the residual diagnostics that motivate the regime structure and the Student- t errors, including the two-step fit of the degrees of freedom.

¹³The ratio lies between zero and one and equals zero before the ON RRP facility became active. It enters in levels, with no log or inverse-hyperbolic transform.

¹⁴The hyperparameters are overall tightness 3, lag decay 0.5, sum-of-coefficients (μ) weight 1, and co-persistence (λ) weight 5. The dummy scales use each variable’s univariate $\text{AR}(p)$ residual standard deviation, and the prior precision stays invariant to units.

3.2 Volatility Regimes and Heavy-Tailed Shocks

For the heteroskedasticity identification I use $h = 4$ volatility regimes. Table 1 reports the dates. Each regime boundary is a boundary of one of the four balance-sheet episodes of Table A.1. The first break is the end of the QE purchases in 2014:M11. The second is the start of the COVID purchases in 2020:M3, and the third is the start of the QT2 runoff in 2022:M6. Each of the four regimes contains one balance-sheet episode, and none contains two.

Table 1: Volatility regimes.

Regime	Start	End	Description
1	2008:M11	2014:M11	QE1 through QE3.
2	2014:M12	2020:M2	Between the QE purchases and the COVID purchases, and QT1.
3	2020:M3	2022:M5	COVID-19 QE.
4	2022:M6	2026:M6	QT2 and the months after it ended.

Note: End dates are inclusive. Each break is a date on the balance sheet, the end of the QE purchases, the start of the COVID purchases, and the start of the QT2 runoff. Each regime contains one balance-sheet episode of Table A.1. The QT2 runoff ended in 2025:M12, and the six months that follow are too few for a regime of their own.

The sample spans the financial crisis, COVID, and the post-2021 tightening. Figure 4 plots the reduced-form residuals u_t in (3) from estimating (1) with a single variance regime and Gaussian errors, without the identification restrictions, under the prior of Section 3.1. Each panel shows the posterior median residual of one equation in its own units. The four subsamples follow the dates of Table 1 and partition the residual sample for description only. The number above each subsample is its residual standard deviation.

The standard deviations vary across subsamples, and they do not move together across equations. The one-year yield is four times as volatile in the third subsample as in the first. The term spread barely moves across the four, and the EBP is less volatile in the last subsample than in the first. Non-proportional patterns of this kind motivate the regime variances and are the condition under which (4) identifies the columns of A_0 .¹⁵

I measure each residual against the typical size of a shock in its own regime, and a few months

¹⁵At the posterior mode of the full model and across the retained draws, the estimated variance ratios are far from mutually proportional (Table D.2 in Appendix D).

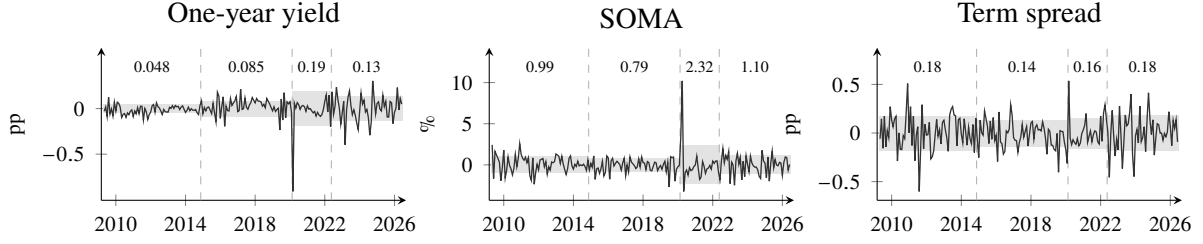


Figure 4: Reduced-form residuals under a single variance regime, three equations.
Note. Each panel plots the posterior median of the reduced-form residual of one equation from the estimation of (1) with a single variance regime, Gaussian errors, no identification restrictions, and the prior of Section 3.1. Units are percent for SOMA and percentage points for the one-year yield and the term spread. Figure D.1 plots the other five equations. Dashed lines mark the regime breaks of Table 1. The shaded band in each subsample is plus and minus one subsample standard deviation of the series, and the number above each subsample is that standard deviation in the units of the panel. The residual sample runs from 2009:M5 to 2026:M6.

still stand far outside the rest. The SOMA residual of April 2020 is 9.6 times that size, and the one-year yield residual of March 2020 is 5.5 times. Those are the months of the emergency purchases at the onset of COVID and of the emergency cuts to the zero lower bound.

Without the latent scales $\nu_{i,t}$ of (4) the estimated variance of a regime that holds one of these months reflects that month rather than the regime. The heteroskedasticity identification weakens with it. The latent scales give each shock in each month its own scale. The two months lose weight in the likelihood, and the regime variances keep their meaning. I fit the Student- t degrees of freedom by the two-step procedure of Brunnermeier et al. (2021) and use the fitted value of 5.84. Section D reports the diagnostics behind these numbers.

3.3 Identification

I identify the shocks with three sets of restrictions. Sign restrictions label five of the eight shocks. Zero restrictions hold at impact. The zeros bind under the average transmission, the responses evaluated at the average of the relative variances across regimes, and need not hold in the regime-specific responses (Appendix C.1). Narrative restrictions tie RFBS and OFBS to the episodes that generated them. All sign restrictions bind from impact through horizon five and must hold both under the average transmission and in every regime. I describe the two balance-sheet shocks first and the three remaining labeled shocks second. Table 2 collects the restrictions on the balance-sheet

shocks, and Table 3 collects the restrictions on the demand, supply, and monetary policy (MP) shocks.

Table 2: Restrictions on the two balance-sheet shocks.

Response variable	RFBS	OFBS
<i>Panel A: sign and zero restrictions</i>		
One-year yield		
SOMA	+	+
log(Reserves + ON RRP)	+	+
ON RRP/(Reserves + ON RRP)	—	+
GDP	Z	Z
GDPDEF	Z	Z
Term spread	—	—
EBP		
<i>Panel B: narrative restrictions (window W, regime average)</i>		
Window W	2020:M4–2021:M2	2023:M9–2024:M7
Mean shock realization, eq. (B.1)	+	—
HD on log(R + RRP), eq. (B.2)	dominant	dominant

Note: A + or – is a sign restriction on the response of the row variable to the column shock, binding from impact through horizon five $[0, 5]$. Z is a zero at impact. Blank cells are unrestricted. Panel B ties each balance-sheet shock to its window. Over the window the mean realization of RFBS must be positive and the mean realization of OFBS must be negative, and the shock’s historical-decomposition contribution to log(Reserves + ON RRP) must exceed the sum of the contributions of all other labeled shocks.

The balance-sheet shocks. Table 2 collects the restrictions on RFBS and OFBS. The two shocks correspond to the two financing channels. I normalize the positive direction of both shocks as an expansion. A positive RFBS shock raises SOMA, raises log(Reserves + ON RRP), and compresses the term spread, and the expansion settles in reserves. A positive OFBS shock moves the same three variables in the same direction, and the expansion settles in ON RRP. The ON RRP ratio separates the two columns. A reserve-financed balance-sheet expansion lowers the ratio (reserves raise the total while ON RRP is unchanged). An ON RRP-financed expansion raises the ratio directly. Without the ratio restriction the two columns would satisfy identical restrictions. The restrictions carry no episode label. QE and QT1 enter as positive and negative realizations of RFBS, and the

QT2 drain enters as negative OFBS realizations.

One economic assumption generates the zero restrictions in Panel A of Table 2.

Assumption 1 (Balance-sheet shocks do not reach real activity and prices within the month). *The two balance-sheet shocks have no impact effect on real GDP or the GDP deflator.*

Assumption 1 follows the timing of [Leeper et al. \(1996\)](#), whose sluggish private sector responds to financial variables only with a delay. Production and final-goods pricing involve planning processes, and movements in interest rates and in the balance sheet reach output and prices smoothly over time rather than within the month. [Christiano et al. \(2005\)](#) impose the same timing on the MP shock. Assumption 1 applies it to the two balance-sheet shocks instead, and the MP shock stays sign-restricted at impact. The balance sheet changes gradually after each program announcement ([Nakashima et al., 2024](#)), and its real effects plausibly arrive after the quantities accumulate. The timing argument binds more tightly for the quantity channel than for the rate channel.

Beyond impact, the GDP and deflator responses to both balance-sheet shocks stay unrestricted. The zeros bind on the average transmission, the response evaluated at the regime average of the relative variances. That average is the object I report. The regime-specific impact responses stay unrestricted ([Section 4](#)). The macroeconomic transmission is data-determined and can differ across the two channels.

I impose two narrative restrictions, one for each balance-sheet shock, and Panel B of Table 2 summarizes them.¹⁶ Each adapts the single-date narrative sign restrictions of [Antolín-Díaz and Rubio-Ramírez \(2018\)](#) to a window average and ties the shock to the episode that generated it through two requirements. The mean realization of the shock over the window must carry the sign of its episode. The shock's contribution to the unexpected movement of $\log(\text{Reserves} + \text{ON RRP})$ over the window must dominate the contributions of the other labeled shocks.¹⁷ The baseline

¹⁶I also estimated the model with four narrative restrictions. The first addition anchors a negative mean RFBS realization over the QT1 window. The second anchors a positive mean OFBS realization over the ON RRP buildup, April 2021 through February 2022. The estimates stay almost identical, and the two restrictions below are enough in this sample.

¹⁷I impose the dominance on $\log(\text{Reserves} + \text{ON RRP})$ rather than on SOMA for two reasons. [Section 2](#) defines the two episodes by their effects on the liability side, and $\log(\text{Reserves} + \text{ON RRP})$ carries no zero restriction. The

measures the contribution by the multi-horizon interval historical decomposition of [Antolín-Díaz and Rubio-Ramírez \(2018\)](#). Appendix [B](#) states the two requirements formally.

I tie the RFBS shock to the first phase of the COVID purchases, April 2020 through February 2021, when the expansion settled predominantly in bank reserves. The window carries a positive sign. From March 2021 the inflows settled increasingly in money market funds and the ON RRP facility (Section [2](#)). Including those months would risk misallocating the TGA-driven movement in $\log(\text{Reserves} + \text{ON RRP})$ to the RFBS shock.

I tie the OFBS shock to the sharp ON RRP drain of September 2023 through July 2024. The window contains eleven months, the same length as the RFBS window. A positive OFBS shock is an expansion, the QT2 drain corresponds to negative OFBS realizations, and the window carries a negative sign. Over that window the ON RRP facility ran down steadily while reserves stayed roughly flat. This window starts more than a year after QT2 began in mid-2022, as Section [2](#) documents.¹⁸ Tying the restrictions to these phases, rather than to full episodes, keeps each inside the window where its channel was the active margin. Averaging over windows rather than single dates makes the restriction robust to the timing noise of individual months.

The demand, supply, and MP shocks. Table [3](#) collects the restrictions on the three remaining labeled shocks. Labeling them serves the narrative restrictions. The dominance requirement compares each balance-sheet shock against the other labeled shocks, and the labels put demand, supply, and MP among the competitors it must dominate over its window. Without the labels the comparison would run against the other balance-sheet shock alone. Demand, supply, and a conventional MP shock follow standard conventions. A demand shock raises the one-year yield,

requirement discriminates the balance-sheet shock against every labeled shock, including the demand, supply, and MP shocks. On SOMA the zero restrictions would leave only the other balance-sheet shock as a competitor.

¹⁸Part of the ON RRP decline over the window coincides with the post-debt-ceiling rebuild of the Treasury General Account (TGA). The Treasury refilled the TGA through heavy bill issuance from June through October 2023, and the rebuild drew down ON RRP rather than reserves. This raises the question of whether to attribute the drain to QT2 or to the fiscal rebuild. The two are far apart in magnitude. From September 2023 onward the monthly decline in ON RRP is roughly three times the contemporaneous rise in the TGA. The rebuild does not account for the bulk of the drain, and among the five labeled shocks OFBS is the dominant driver of $\log(\text{Reserves} + \text{ON RRP})$ over the window. I read the restriction as identifying the dominant balance-sheet shock, while noting that the bill issuance associated with the rebuild plausibly amplified the same channel. Section [5.2](#) reports specifications that control the TGA and Treasury issuance directly.

GDP, and the GDP deflator. A supply shock raises GDP and lowers the deflator. An MP shock raises the one-year yield and lowers GDP and the deflator. One economic assumption generates the zero restrictions on SOMA.

Table 3: Restrictions on the demand, supply, and MP shocks.

Response variable	Demand	Supply	MP
One-year yield	+		+
SOMA	Z	Z	Z
log(Reserves + ON RRP)			
ON RRP/(Reserves + ON RRP)			
GDP	+	+	—
GDPDEF	+	—	—
Term spread			
EBP			

Note: A + or — is a sign restriction on the response of the row variable to the column shock, binding from impact through horizon five $[0, 5]$. Z is a zero at impact. Blank cells are unrestricted. The regime covariances identify (A_0, Λ) and do not label the shocks, and the three unlabeled shocks carry no restriction.

Assumption 2 (Balance-sheet policy drives SOMA within the month). *Within the month the Federal Reserve moves its SOMA holdings only through its balance-sheet policy. The demand, supply, and MP shocks have no impact effect on SOMA.*

Assumption 2 sets three zeros on SOMA, and the grounds differ by shock. For the demand and supply shocks the zero follows the policy-equation exclusions of [Leeper et al. \(1996\)](#). In their system the policy instrument does not respond within the month to output and final-goods prices, the counterparts of GDP and the GDP deflator here. They give two grounds. The data for those series arrive only after the month ends, and a shift in the policy stance requires consultation and consensus building. The preannounced schedules and caps of asset purchases and runoff strengthen both grounds, and the within-month path of the balance sheet does not react to macroeconomic news. The restriction constrains the impact responses of SOMA only, and GDP and the GDP deflator carry zeros under Assumption 1 instead.

For the MP shock the zero reflects the separation of the two instruments. Since October 2008 the Federal Reserve has implemented the policy rate through administered rates on reserves and the ON RRP facility rather than through open market operations. A rate surprise requires no change in asset holdings within the month. An MP shock that moved SOMA contemporaneously would also conflate the two instruments. The sample begins in November 2008 and lies entirely within this operating framework.

[Leeper et al. \(1996\)](#) find that movements in reserves and M1 in their sample arise mostly from policy accommodating shifts in private demand. An identification that treats reserves or M1 as a policy instrument mislabels those demand shifts as policy shocks. They call such identifications unreliable. The critique applies to any quantity that private demand can move. SOMA is not such a quantity. Only the Federal Reserve trades in its own portfolio, and under administered rates the accommodation of demand runs through the liability side rather than through asset holdings.

4 HARS-Z

This section presents the sampler that draws from the posterior of the heteroskedastic model of [Section 3](#) conditional on the sign, zero, and narrative restrictions of [Section 3.3](#). HARS-Z is the heteroskedastic absorb-and-rotate sampler with zero restrictions. I develop it by extending HARS, the heteroskedastic absorb-and-rotate sampler of [Kim and Zha \(2026\)](#), hereafter KZ) that combines shock heteroskedasticity with sign and narrative restrictions, to include zero restrictions. I adopt the notation of KZ throughout. [Appendix C](#) contains the full theoretical analysis. Readers focused on the estimates can note [Algorithm 1](#) and continue to [Section 5](#).

Throughout, $\theta = (A_0, \Lambda, A_+)$ collects the structural objects with $\Lambda = \{\Lambda_k\}_{k=1}^h$, $Q \in \mathcal{O}(n)$ denotes the rotation, and $SR(\theta, Q)$ is the indicator that the sign and narrative restrictions hold. HARS (KZ, [Algorithm 1](#)) combines likelihood-preserving posterior draws of the model parameters with rotations drawn from a well-defined probability distribution restricted by the sign and narrative restrictions, adopting a column-wise elliptical slice sampler (ESS; [Murray et al., 2010](#)). Sign and

narrative restrictions are inequalities. They truncate the conditional distribution of Q to a subset of positive Haar measure, and the posterior target in HARS (Appendix C.1) is well defined. A random rotation lands in the subset with positive probability (Figure 5, panel a). Appendix C.1 reviews the sampler and the repair steps in detail.

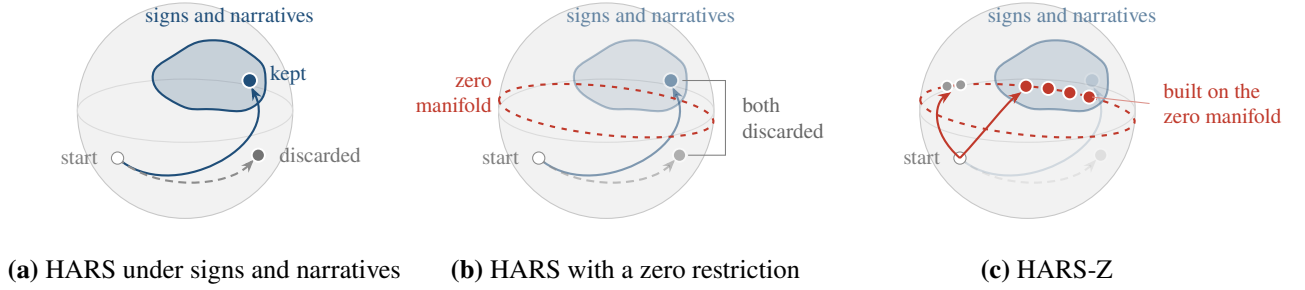


Figure 5: Why zero restrictions break HARS, and what HARS-Z does instead.

Note. Each sphere stands for $O(n)$. The shaded region is the set of rotations that satisfy the sign and narrative restrictions, and the dashed curve is the zero manifold $Q(\theta)$. In panel (a) HARS draws rotations from the starting point and keeps the ones that land in the region. In panel (b) a zero restriction adds the curve, a set of measure zero. A random rotation reaches it with probability zero, and every draw is discarded. In panel (c) HARS-Z builds each rotation on the curve and the region selects among them.

A zero restriction is not a truncation. A zero restriction on the impulse responses is a linear equality constraint on the j -th column q_j of Q . Following the notation of Arias et al. (2018, hereafter ARR), the $z_j \times n$ matrix $Z_j(\theta)$ stacks the z_j zero restrictions on shock j as rows. The restrictions on shock j read $Z_j(\theta) q_j = 0$, the rows depend on the draw through A_0 and the moving-average coefficients, and the sampler must rebuild them at every iteration. The zeros bind under the average transmission only and need not hold in the regime-specific responses, where $\Lambda_k^{1/2}$ reweights the constrained column. Define the zero manifold

$$Q(\theta) = \{ Q \in O(n) : Z_j(\theta) q_j = 0 \text{ for every restricted } j \}. \quad (5)$$

Whenever some $z_j \geq 1$, $Q(\theta)$ is a lower-dimensional smooth manifold in $O(n)$ and $\mu(Q(\theta)) = 0$, where μ is Haar measure on $O(n)$, as ARR note in motivating their Algorithm 2 (Figure 5, panel b).

One may be tempted to impose the zeros by adding them to $SR(\theta, Q)$ and running HARS

unchanged. The zeros lower the dimension of the admissible set $R(\theta) = \{Q \in O(n) : SR(\theta, Q) > 0\}$ instead of truncating it, and the condition $\mu(R(\theta)) > 0$ fails at every θ with at least one zero restriction. The restricted posterior target is not defined, the Gaussian coordinatization never reaches the zeros, and every step of KZ’s Algorithm 1 fails with probability one, including its initialization. The failure originates in the measure rather than in the algorithm. Appendix C.2 shows it formally. The sampler must parameterize the zero-restricted set directly rather than reach it by sampling. It must build the rotation on $Q(\theta)$, and its slice-sampling updates must never leave it (Figure 5, panel c). The rest of this section constructs exactly such a parameterization.

HARS-Z replaces the QR map γ with a map $\gamma_z(\cdot; \theta)$, built on the null-space construction of ARRW’s Algorithm 2, that sends a Gaussian matrix to a rotation on $Q(\theta)$ by construction. These are the built rotations of Figure 5, panel c. The state of the chain is the pair (θ, X) , where X is an $n \times n$ Gaussian matrix, the coordinates in which HARS runs its column-wise ESS update. The sampler recomputes the rotation $Q = \gamma_z(X; \theta)$ whenever it needs one. A structural update moves the manifold $Q(\theta)$, the recomputed rotation lies on the new manifold, and the zeros hold after every move. Appendix C.3 builds the map γ_z and establishes its properties.

HARS-Z targets the analogue of the posterior target in HARS on the zero-restricted set, with the uniform measure on $Q(\theta)$ in place of Haar measure on $O(n)$,

$$\pi_v(d\theta, dQ \mid y, SR) = \pi(\theta \mid y) \frac{\mathbf{1}\{Q \in R(\theta)\} \mu(dQ \mid Q(\theta))}{\mu(R(\theta) \mid Q(\theta))} d\theta, \quad (6)$$

where $\mu(\cdot \mid Q(\theta))$ denotes the uniform measure on $Q(\theta)$ and $R(\theta) = \{Q \in Q(\theta) : SR(\theta, Q) > 0\}$ now denotes the admissible set on the zero-restricted set. Generating the rotation through γ_z induces the null-space construction measure on $Q(\theta)$, not the uniform measure. Weighting each retained rotation by the volume element $v(Q; \theta)$ of the construction, normalized within each retained draw of θ , recovers the target π_v . The column-wise ESS update remains valid on $Q(\theta)$ and leaves the conditional target invariant. Appendix C.4 establishes these results.

The HARS-Z sampler, stated below as Algorithm 1, alternates a posterior draw of θ with a

rotation draw of X . The posterior draw is the sampler of Brunnermeier et al. (2021, hereafter BPSS), taken from KZ unchanged. Following KZ’s Theorem 1, partition the parameter collection as $z_0 = (A_0, \Lambda)$ and $z_+ = A_+$, and let s_0 collect everything held fixed in the block update. The first block moves by Metropolis–Hastings (MH) on the collapsed density with A_+ integrated out, and the companion block moves by a draw from its Gibbs conditional $p_+(A_+ | z_0, s_0)$. The rejection counts generated by the repair steps below are diagnostics of computational efficiency, not density terms (KZ, Proposition 5). Each repair returns a draw from the correct conditional distribution. The counts require no reweighting and enter no posterior computation. Appendices C.3 and C.4 state the lemma, propositions, and corollaries that Algorithm 1 cites.

Algorithm 1. (Heteroskedastic absorb-and-rotate sampler with zero restrictions (HARS-Z).)

Let the current admissible state be (θ, X) with $Q = \gamma_z(X; \theta)$ and $SR(\theta, Q) > 0$, and fix the tuning parameters $N_{\text{ess}}, N_\theta, N_Q$. For the initialization, construct an admissible rotation by a numerical search at the initial θ and convert it into Gaussian coordinates by Lemma 1(iii). One iteration proceeds as follows. Each step matches its counterpart in Algorithm 1 of KZ.

1. **Posterior draw with admissibility check.** Propose $z_0^{\text{prop}} = (A_0^{\text{prop}}, \Lambda^{\text{prop}})$ by the first-block MH step and set $z_0^\star = z_0^{\text{prop}}$ if accepted and $z_0^\star = z_0$ otherwise. Draw the companion block $z_+^\star \sim p_+(A_+ | z_0^\star, s_0)$ together with the latent Student- t scales, and form the completed draw $\theta^\star = (z_0^\star, z_+^\star)$. Rebuild the rotation at the new draw, $Q^\star = \gamma_z(X; \theta^\star)$. The rebuilt rotation satisfies every zero restriction by construction. Check the sign and narrative restrictions. If $SR(\theta^\star, Q^\star) > 0$, set $\theta \leftarrow \theta^\star$ and go to Step 3. Otherwise the rotation is stale. Go to Step 2a.
- 2a. **Posterior-draw repair by the joint procedure.** Repair on the completed draw, as required by KZ’s Theorem 1. For at most N_θ attempts, draw a completed candidate $\theta^{\text{cand}} = (z_0^{\text{cand}}, z_+^{\text{cand}})$ and retain it only if $SR(\theta^{\text{cand}}, \gamma_z(X; \theta^{\text{cand}})) > 0$. The first-block MH step, anchored at the last admissible state θ (not at the stale draw $\theta^\star$), proposes the candidate block z_0^{cand} , and $z_+^{\text{cand}} \sim p_+(A_+ | z_0^{\text{cand}}, s_0)$. On success, set $\theta \leftarrow \theta^{\text{cand}}$ and go to Step 3. If the N_θ attempts retain no candidate, discard all candidates and go to Step 2b, carrying the draw $\theta^\star$ that

triggered the repair. Failed candidates, and the stale draw $\theta^\star$ itself, are auxiliary proposals. The chain never retains them.

- 2b. **Global rotation repair.** Hold $\theta^\star$ fixed. For $j = 1, \dots, N_Q$, draw a fresh Gaussian matrix $G^{(j)}$ and propose $Q^{(j)} = \gamma_z(G^{(j)}; \theta^\star)$, the analogue on $Q(\theta^\star)$ of the independent Haar draws in Step 2b of KZ’s Algorithm 1. Let $j^\star$ be the first index with $SR(\theta^\star, Q^{(j^\star)}) > 0$. If such an index exists, set $(\theta, X) \leftarrow (\theta^\star, G^{(j^\star)})$ and go to Step 3. If no index succeeds after N_Q attempts, declare sampler failure and proceed to Step 5.
3. **Rotation draw.** From the admissible state, run N_{ess} column-wise ESS updates of X at fixed θ (Proposition 2), producing Gaussian states $X^{(1)}, \dots, X^{(N_{\text{ess}})}$ and rotations $Q^{(r)} = \gamma_z(X^{(r)}; \theta)$ with $SR(\theta, Q^{(r)}) > 0$ for every r . Retain all N_{ess} rotations together with θ for posterior impulse-response computation.
4. **Advance the state.** Set the next state to $(\theta, X^{(N_{\text{ess}})})$ and return to Step 1.
5. **Sampler failure.** If a posterior move produces a draw $\theta^\star$ for which neither the joint posterior-draw repair nor the global rotation repair restores sign admissibility, the sampler cannot advance. Increase N_θ or N_Q , improve the repair search, or revisit the restriction set.

Retaining N_{ess} rotations per structural draw divides the cost of the structural update across several rotation draws. The ESS updates evaluate only the admissibility indicator, not the likelihood. Appendix C.4 establishes the validity of Algorithm 1 step by step. Posterior inference weights each retained rotation by its volume element, normalized within each retained draw of θ , and reports weighted medians and bands. Appendix C.5 states the estimator. HARS-Z nests HARS. With an empty zero set, γ_z reduces to the QR map γ , $Q(\theta) = O(n)$, and the construction measure $\mu_z(\cdot; \theta)$ is Haar measure. The volume weight is identically one, the weighted summaries reduce to the unweighted ones, and the sampler reduces to KZ’s Algorithm 1.

5 Results

I estimate the baseline posterior with HARS-Z. Appendix D reports the sampler settings and the diagnostics.

5.1 Baseline Impulse Responses

Figures 6 and 7 report the impulse responses to the two balance-sheet shocks under the baseline identification of Section 3.3, with the relative variances set to their average of one across regimes. The sign and narrative restrictions of Section 3.3 define each shock in its expansionary direction, and the identification is unchanged. Each figure shows one shock in its contractionary direction, the direction of the QT episodes. To make the two channels directly comparable, I normalize each shock to a one-percent impact decline in SOMA holdings. The decline continues after impact. Figure E.1 in Appendix E reports the responses to the demand, supply, and MP shocks.

Figure 6 plots the responses to the RFBS shock. The shaded regions are the 68% credible bands. The restricted responses hold by construction. Panel A of Table 2 restricts the expansionary direction. Under the contractionary display, SOMA, $\log(\text{Reserves} + \text{ON RRP})$, the ON RRP ratio, and the term spread move with the signs of Panel A reversed. The zero restrictions hold GDP and the GDP deflator at zero. The economic content lies in the GDP and GDP-deflator responses beyond impact and in the EBP response, where no restriction binds.

A negative RFBS shock lowers GDP persistently. The GDP response turns negative at five months and falls through the horizon, and the deflator response turns negative at two months. The 68% bands exclude zero from eleven months onward for GDP and from nineteen months onward for the deflator. The same shock raises the EBP. The response rises through the first year and a half and stays positive through the horizon, and the 68% bands exclude zero from two months onward.

Figure 7 plots the responses to the OFBS shock. The restricted responses again hold by construction and reverse the signs of the OFBS column in Panel A of Table 2. The economic content again lies in the GDP and GDP-deflator responses beyond impact and in the EBP response, where

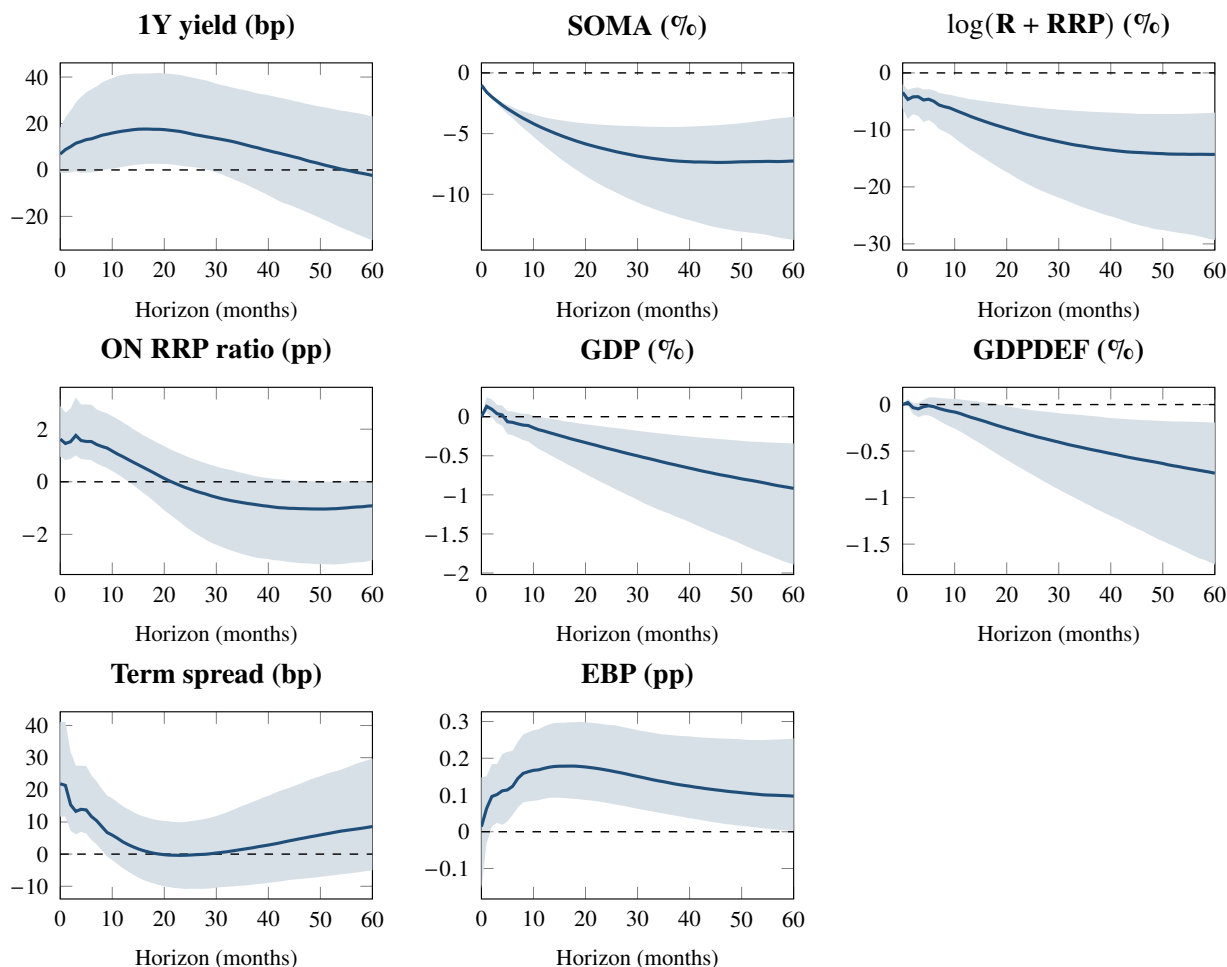


Figure 6: Impulse responses to the RFBS shock.

Note. Impulse responses to the RFBS shock, normalized to a one-percent impact decline in SOMA holdings. I evaluate the responses at the regime average of the relative variances. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. Vertical axes are in basis points for the one-year yield and the term spread, percent for SOMA, $\log(\text{Reserves} + \text{ON RRP})$, GDP, and GDPDEF, and percentage points for the ON RRP ratio and the EBP.

no restriction binds.

A negative OFBS shock lowers the deflator. The response turns negative at one month and falls through the horizon, and the 68% bands exclude zero from six months onward. The GDP bands include zero beyond two months, and the EBP bands include zero at every horizon.

Interpreting the two shocks. Neither figure describes a single episode. The linear model has one response function for each shock, and realizations of both signs enter its estimation. QE and QT1 enter the RFBS shock as positive and negative realizations, and the ON RRP buildup and the QT2

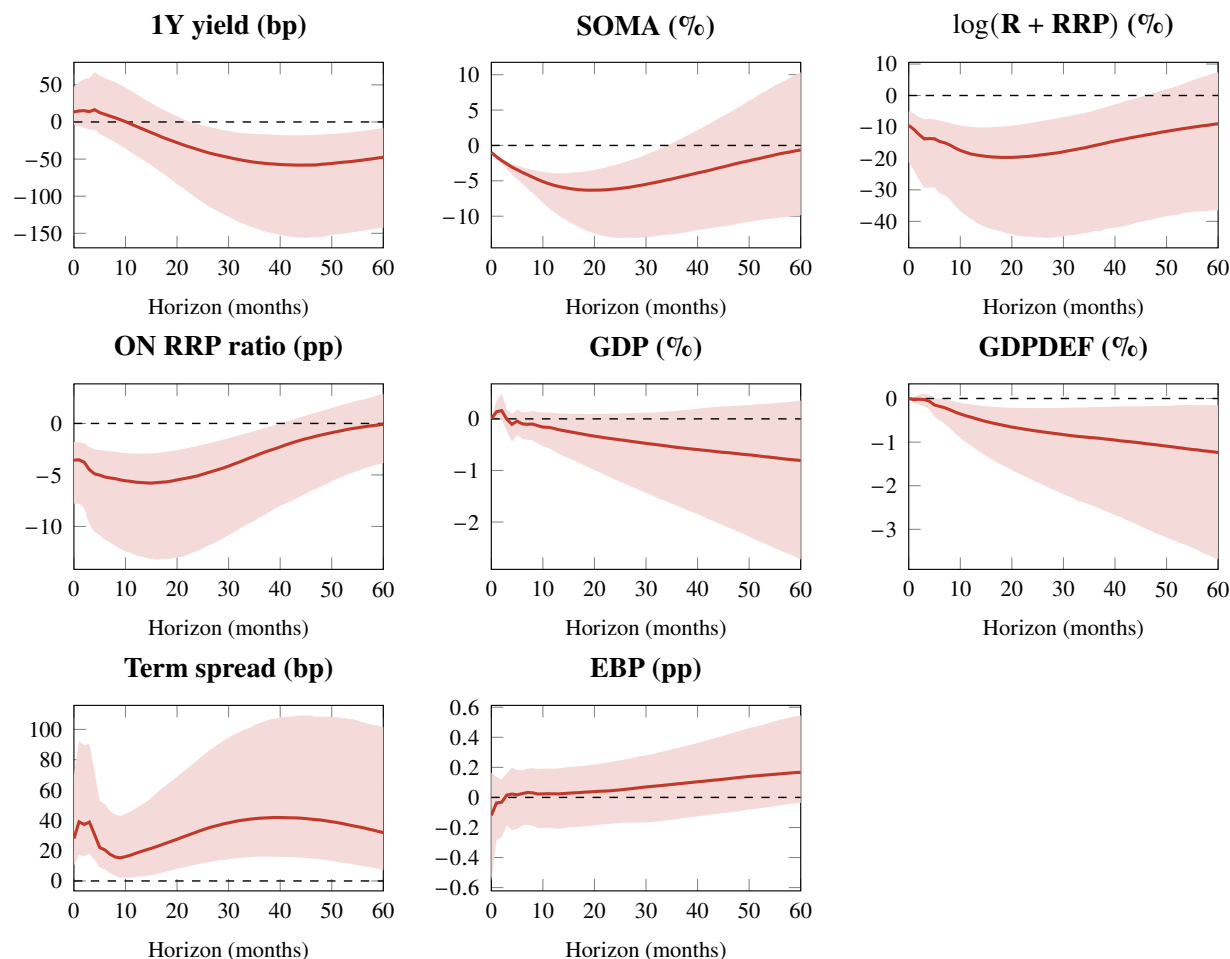


Figure 7: Impulse responses to the OFBS shock.

Note. Impulse responses to the OFBS shock, normalized to a one-percent impact decline in SOMA holdings. The figure shows the shock in the direction of the QT2 drain. I evaluate the responses at the regime average of the relative variances. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. Vertical axes are in basis points for the one-year yield and the term spread, percent for SOMA, $\log(\text{Reserves} + \text{ON RRP})$, GDP, and GDPDEF, and percentage points for the ON RRP ratio and the EBP.

drain enter the OFBS shock the same way (Section 3.3). Realizations of both signs also fall outside those episodes. The contraction display is a sign convention. I read each plotted response as the average over both directions and not as the response to the QT episode alone. In what follows I call a response present only where its 68% band excludes zero.

Only the reserve-financed operation has real effects, and the EBP says why. A reserve-financed operation moves the EBP, a contraction raising it and an expansion lowering it, and an ON RRP-financed operation does not move the EBP. This is the prediction of Section 3.1. Interest on reserves is bank income and bank capital, and banks with more capital lend at a smaller spread. A

reserve-financed operation moves that spread, and GDP moves with it, a decline after a rise in the EBP and a rise after a decline, as in [Gilchrist and Zakrajšek \(2012\)](#). The ON RRP rate builds no capital anywhere, and no real effect follows.

5.2 Robustness

Nine specifications each change one element of the baseline of [Section 5.1](#) and keep every other element. Three remove zero restrictions, [Assumption 1](#), [Assumption 2](#), and both. Three change the variance structure. They use one regime variance (homoskedasticity) in place of four, and Gaussian errors in place of the Student- t with latent scales. The third sets the regime breaks at the exits from and the return to the zero lower bound (2015:M11, 2020:M3, and 2022:M2) in place of the balance-sheet episodes. Three change the regressors. Two add exogenous controls, the TGA ratio in one and the bill share and coupon issuance in the other. The third puts industrial production and the CPI in place of the interpolated GDP and deflator. [Table 4](#) gives, for each specification, the month from which the 68% bands exclude zero for the responses of GDP, the deflator, and the EBP to each of the two shocks. [Appendix F](#) plots those responses.

The main results survive every change. In all nine specifications a reserve-financed contraction lowers GDP and raises the EBP, and an ON RRP-financed contraction lowers the deflator and moves neither GDP nor the EBP. The specifications differ only in when each response appears, and [Table 4](#) gives the month.

Two things change. The reserve-financed decline in the deflator is the fragile result. It disappears when the impact zeros on GDP and the deflator are dropped and when the CPI replaces the deflator. Without the impact zeros the RFBS shock also carries a positive GDP response at impact, and the decline appears in the third year. I read that positive response as the state of the economy around the balance-sheet actions, the COVID months in the main, rather than as transmission. The impact zeros are what separates the two. The other change is smaller. With Treasury issuance as a control, the RFBS rise in the EBP and the OFBS decline in the deflator last about two years instead of the full horizon. Bill issuance is the route the ON RRP drain takes to the Treasury ([Section 2](#)), and a

Table 4: Robustness of the GDP, deflator, and EBP responses.

Specification	RFBS			OFBS		
	GDP	Deflator	EBP	GDP	Deflator	EBP
Baseline	11	19	2	none	6	none
Without Assumption 1	33	none	3	none	13	none
Without Assumption 2	11	16	2	none	6	none
Without any zero	34	none	3 to 46	none	12	none
One variance regime	35	24	8	none	9	none
Gaussian errors	23	27	7	none	6	none
Regime breaks at the lower bound	15	25	6	none	7	none
TGA control	6	28	2	none	8	none
Issuance controls	9	12	2 to 29	none	9 to 28	none
Industrial production and CPI	17	none	3 to 54	none	5	none

Note: Each cell is the first month, counting from impact as month zero, at which the 68% band excludes zero with the sign the baseline response has. That sign is a decline for GDP and the deflator and a rise for the EBP. The band excludes zero through month sixty unless a second month closes the range. “none” means the band excludes zero with that sign at no horizon. Under the industrial-production specification the GDP and deflator columns read industrial production and the CPI. Without any zero the OFBS EBP band excludes zero at month sixty alone. In the baseline and in five specifications the OFBS GDP band excludes zero with a positive median over the first one to two months and at no later horizon.

control that absorbs the route absorbs part of the OFBS shock.

5.3 Variance and Historical Decompositions

The impulse responses report an elasticity for a one-percent decline in SOMA holdings. They do not say how much of the observed variation in GDP and the deflator the two shocks account for. This subsection reports two decompositions that answer that question. The forecast-error variance decomposition gives the share of the forecast variance at a given horizon. The historical decomposition gives the contribution of each shock to the unexpected change in a variable over an episode. Both use the baseline draws of [Section 5.1](#) and weight the draws as the impulse responses do.

Forecast-error variance decomposition. [Table 5](#) reports the variance decomposition under the average transmission. The shares below are posterior medians at three years, and the table gives the

68% intervals around them. The two balance-sheet shocks together account for 10% of the forecast variance of GDP and 11% of the forecast variance of the deflator. Of the GDP forecast variance the RFBS shock accounts for 7% and the OFBS shock for 3%. Of the deflator forecast variance the RFBS shock accounts for 5% and the OFBS shock for 6%.

The EBP is the variable the two shocks account for most, at 22% together, and the RFBS shock alone accounts for 18% of it. That is the largest share any labeled shock takes in any row of [Table 5](#).

Table 5: Forecast-error variance decomposition.

	Horizon	Share of forecast-error variance, percent				
		Demand	Supply	MP	RFBS	OFBS
GDP	12	10.2	11.7	10.1	1.4	0.8
		[4.6, 20.7]	[4.5, 23.9]	[4.2, 20.9]	[0.5, 3.7]	[0.3, 2.4]
	24	7.1	11.8	11.6	3.8	1.7
		[3.2, 16.3]	[3.9, 26.1]	[4.2, 24.2]	[1.0, 11.2]	[0.4, 6.1]
	36	6.0	11.0	11.8	6.9	2.8
		[2.6, 14.4]	[3.2, 27.1]	[3.7, 26.4]	[1.5, 17.9]	[0.5, 10.0]
GDPDEF	12	14.2	28.8	9.1	0.9	2.0
		[3.5, 36.2]	[8.0, 55.2]	[2.8, 23.5]	[0.2, 3.4]	[0.4, 5.5]
	24	9.5	26.5	10.8	2.6	4.7
		[2.3, 28.0]	[7.6, 50.3]	[2.7, 27.0]	[0.5, 9.3]	[0.9, 11.7]
	36	7.4	23.4	11.1	4.7	6.1
		[2.1, 23.4]	[6.4, 45.8]	[2.5, 28.4]	[0.8, 15.3]	[1.1, 16.0]
EBP	12	5.9	6.4	9.1	16.4	4.6
		[1.9, 15.6]	[1.8, 20.1]	[2.5, 26.4]	[5.7, 37.3]	[1.4, 13.6]
	24	5.4	5.9	7.7	18.9	3.8
		[1.6, 15.1]	[1.5, 19.8]	[2.3, 21.8]	[5.7, 39.4]	[1.1, 11.7]
	36	5.4	6.0	7.4	17.5	4.2
		[1.6, 16.1]	[1.5, 19.9]	[2.3, 20.9]	[5.0, 38.0]	[1.2, 12.2]

Notes: Shares of the forecast-error variance in percent, under the average transmission, with the relative variances at their average of one across regimes. Horizons are in months. The first row of each pair is the posterior median and the second is the 68% credible interval, both over the retained draws. The five labeled shocks appear here, and the three unlabeled shocks account for the remainder.

Historical decomposition. The historical decomposition splits the accumulated forecast error of a variable over a window among the shocks that the sample realizes inside it. Let $W = \{t_0 + 1, \dots, t_1\}$

and let $\mathcal{I}_{t_0}$ collect the observables through t_0 . The contribution of labeled shock j to variable i is

$$\text{HD}_{j \rightarrow i}(W) = \sum_{\ell=0}^{t_1-t_0-1} [\Theta_{s_{t_1-\ell}, \ell}(Q)]_{ji} \tilde{\varepsilon}_{j, t_1-\ell}, \quad \sum_{j=1}^n \text{HD}_{j \rightarrow i}(W) = y_{i, t_1} - \mathbb{E}[y_{i, t_1} | \mathcal{I}_{t_0}], \quad (7)$$

where the regime index follows the date of the shock. [Figure 8](#) reports (7) over the four volatility regimes and over the two halves of the fourth, as posterior medians with 68% intervals, for the five labeled shocks. The split is at 2023:M5, and the narrative window of the OFBS shock lies inside the second half ([Section 3.3](#)). [Figure G.1](#) in [Appendix G](#) reports the same decomposition for the other five variables.

Reserves financed the balance sheet over the first two regimes ([Figure 2](#)), and the RFBS shock is the largest contributor to the GDP, deflator, and EBP forecast errors in both. Over Regime 1, the QE years, it pushes GDP and the deflator above their forecasts and the EBP below. Over Regime 2, which holds QT1, it pulls GDP and the deflator below their forecasts and pushes the EBP above. The OFBS shock ranks last or near last in both regimes.

Over Regime 3 the ON RRP balances build up to near \$2 trillion ([Figure 2](#)) while the COVID purchases add reserves as well. The supply shock is the largest contributor to the GDP and deflator forecast errors. The RFBS shock is the largest contributor to the EBP forecast error. It moves all three variables as it did over Regime 1, GDP and the deflator up and the EBP down. The OFBS shock ranks second for the deflator, which it pushes above its forecast, and contributes little to GDP.

I split Regime 4 at 2023:M5, the month before the ON RRP balances began to fall, to separate the year of QT2 in which the balances barely move from the months in which they fall. Over Regime 4 as a whole, no labeled shock has a band that excludes zero for GDP or the deflator. Over the first year the MP shock is the largest contributor to the GDP and deflator forecast errors, and neither band excludes zero. Over the drain the OFBS shock is the largest contributor to the deflator and GDP forecast errors. It pulls the deflator below its forecast, and the band excludes zero. It pulls GDP below its forecast, and the band includes zero. The RFBS shock contributes nothing to either variable over the drain.

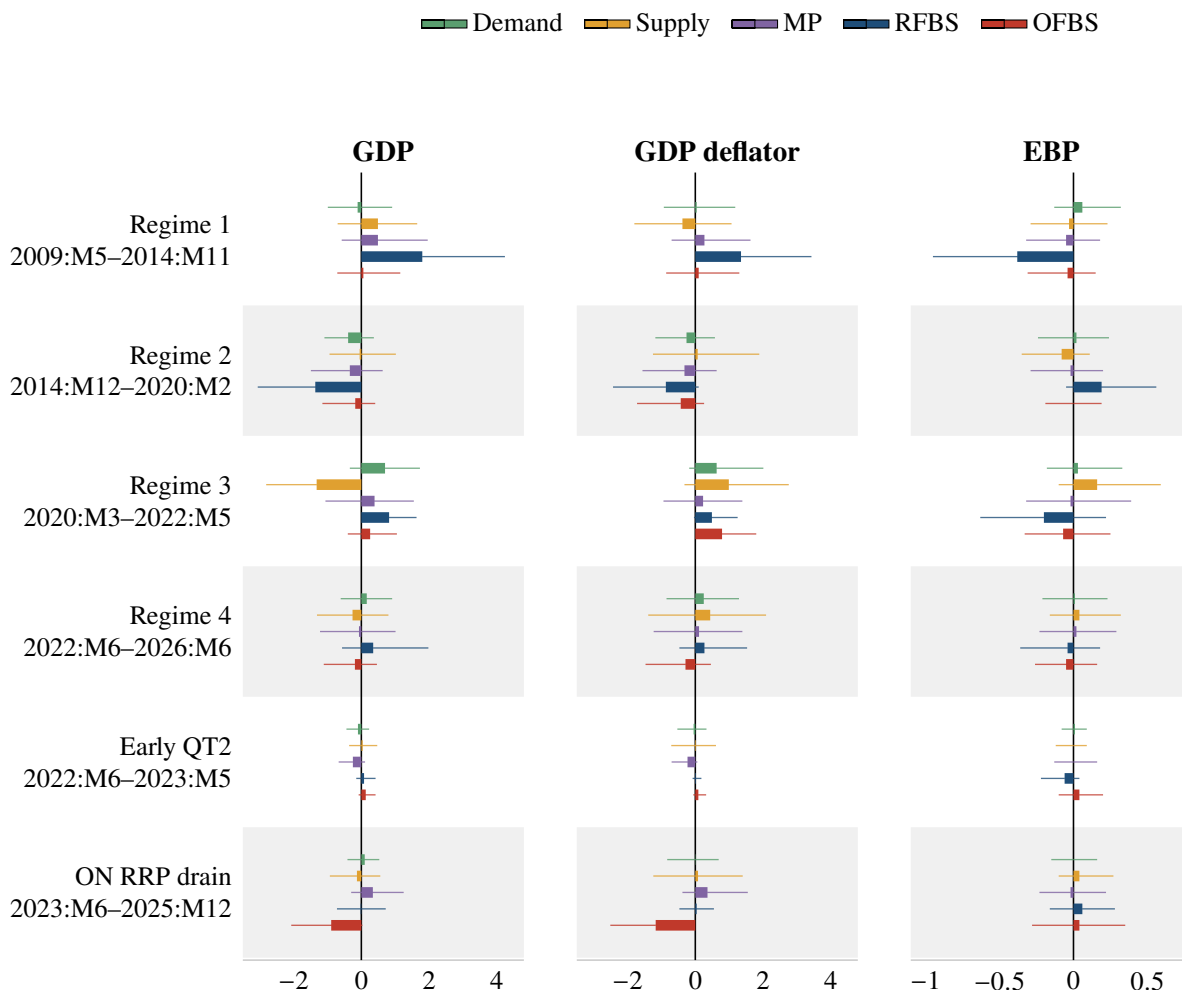


Figure 8: Historical decomposition by volatility regime.

Note. Contribution of each labeled shock to the accumulated forecast error of the variable over the window. Units are percent for GDP and the deflator and percentage points for the EBP. Bars are posterior medians and whiskers are 68% credible intervals, both over the retained draws. The first four windows are the volatility regimes of Table 1, and the last two split the fourth at 2023:M5. Regime 1 opens at 2009:M5, the first estimable month under the six estimation lags. The three unlabeled shocks account for the remainder.

Interpreting the decompositions. The decompositions place each shock in the regimes its liability financed. The RFBS shock moved GDP, the deflator, and the EBP in the three regimes that reserves financed. GDP and the deflator rose when the EBP fell and fell when it rose, as [Section 2](#) predicts for an operation that works through the spread. The OFBS shock moved the deflator over the ON RRP drain and did not move GDP. The drain took the ON RRP rate away from money market funds. As [Section 2](#) predicts, it did not move the EBP, and output did not respond.

The shares in [Table 5](#) say where the financing matters. The two shocks account for a tenth of the

forecast variance of GDP and of the deflator, a small part next to the supply and demand shocks. They account for a fifth of the forecast variance of the EBP, the largest share any labeled shock takes in any variable. The financing of the balance sheet is a second-order factor for output and prices and the first-order factor for the credit spread, which is where [Section 2](#) puts it.

6 Conclusion

Since 2008 the Federal Reserve has financed its balance-sheet operations with two overnight liabilities. Reserves absorbed the purchases after the financial crisis and paid for QT1. ON RRP absorbed much of the COVID purchases and paid for QT2, which left reserves nearly intact ([Figure 2](#)). Both liabilities are overnight claims on the Federal Reserve, and the interest on each goes to a different holder. Interest on reserves is bank income and adds to bank capital. The ON RRP rate passes through money market funds to their shareholders and builds no capital on the way. This paper asks whether the transmission of balance-sheet policy to real GDP and the price level depends on which claim finances it.

Both balance-sheet contractions lower the deflator, and only the reserve-financed contraction lowers GDP. The EBP says why. A reserve-financed operation moves the EBP, a contraction raising it and an expansion lowering it, and an ON RRP-financed operation does not move the EBP. The two balance-sheet shocks account for a tenth of the forecast variance of GDP and the deflator and a fifth of that of the EBP, mostly through the RFBS shock. In the sample the RFBS shock is the largest contributor to the GDP, deflator, and EBP forecast errors over the two regimes that reserves financed. The OFBS shock is the largest contributor to the deflator forecast error over the ON RRP drain of QT2. The results survive nine changes to the specification.

The empirical design combines heteroskedasticity, sign restrictions, narrative restrictions, and zero restrictions. The heteroskedastic likelihood separates the model shocks, and the sign and narrative restrictions label the two balance-sheet shocks. HARS-Z imposes the impact-zero restrictions on that combination, and the rotation satisfies every zero restriction at every draw.

Which liability finances an operation decides whether the operation reaches output at all. Both financings move the price level, and only the one that pays interest to banks moves output. The Federal Reserve sets the terms that govern the split between reserves and ON RRP. Within those terms the composition of Treasury issuance, bank regulation, and the portfolio choices of MMFs affected which liability each episode filled or drained ([Section 2](#)). The asset side of the balance sheet moves the same way in every episode, and the liability side does not. The size of the balance sheet does not summarize what a balance-sheet policy does.

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A Balance-Sheet Data

This appendix collects the episode-level balance-sheet evidence discussed in Section 2. Table A.1 reports the Federal Reserve’s flows by episode. Table A.2 reports the money market fund and hedge fund balance sheets over the same windows, and Figure A.1 reports who held the ON RRP balance.

Table A.1: Federal Reserve balance-sheet flows by episode, USD billions.

	QE 2008-12 → 2014-11	QT1 2017-10 → 2019-08	COVID QE 2020-03 → 2022-03	QT2 2022-06 → 2025-12
<i>Asset side</i>				
Total assets	+2,266	−690	+4,315	−2,352
MBS	+1,725	−267	+1,340	−666
SOMA Treasury	+1,975	−375	+2,732	−1,560
notes/bonds	+1,993	−378	+2,732	−1,467
T-bills	−18	+3	0	−92
<i>Liability side</i>				
Total liabilities	+2,266	−690	+4,315	−2,352
Reserves	+1,765	−709	+1,998	−312
ON RRP	+107	−115	+1,588	−2,152
TGA	+12	−43	+242	+124
Other	+382	+177	+487	−12
<i>Absorption ratios (% of total asset change)</i>				
Reserves	77.9	102.8	46.3	13.3
ON RRP	4.7	16.7	36.8	91.5

Notes: Asset-side figures use the Federal Reserve’s H.4.1 series (WALCL) and the New York Fed’s monthly SOMA Treasury holdings by maturity. Liability-side figures use H.4.1 series (WRESBAL, WTREGEN, WCURCIR) and the daily ON RRP series (RRPONTSYD). “Other” aggregates currency in circulation and residual Federal Reserve liabilities (capital, foreign and other deposits). Total liabilities equal total assets by accounting identity. All values resampled to month-start.

Table A.2 records the ON RRP route. When money market funds take cash out of the facility, the offsetting item sits on their own asset side, in Treasury securities and in repo with private borrowers. Total assets need not fall. The hedge fund rows record who borrowed the repo cash and what they held. The issuance rows record the coupons and bills that the Treasury sold over each window, which is the supply the funds and the hedge funds absorbed.

Figure A.1 records who held the ON RRP balance. The fund categories are the funds’ own Form N-MFP categories, pooled as the Office of Financial Research pools them, with Treasury,

Table A.2: Money market fund and hedge fund balance sheets by episode, USD billions.

	QE	QT1	COVID QE	QT2
	2008-12 → 2014-11	2017-10 → 2019-08	2020-03 → 2022-03	2022-06 → 2025-12
<i>ON RRP facility</i>				
All counterparties	+149	−304	+1,870	−1,873
of which MMFs		−286	+1,656	−1,681
<i>Money market fund assets</i>				
Treasury securities		+222	+719	+1,979
Private repo		+609	−605	+2,251
of which through FICC		+173	−106	+1,210
Total assets		+803	+1,058	+3,316
<i>Hedge funds</i>				
Government debt exposure		+953	−493	+2,497
Repo borrowing		+610	−212	+2,289
<i>Treasury issuance</i>				
Net bill issuance	−564	+530	+1,364	+2,874
Net coupon issuance	+7,163	+1,419	+5,006	+4,092

Notes: Changes over the episode windows of Table A.1. Money market fund, facility, and issuance rows are month-end levels at the last month of the window less the level at the last month-end before it. Form N-MFP is a month-end snapshot. Hedge fund rows are quarter-ends that bracket the window. Blank cells have no observation at the base date. Form N-MFP begins in 2010:M11 and Form PF in 2013, both after the QE window opens. MMFs are money market funds. Private repo is their repo with every counterparty other than the Federal Reserve. FICC is the Fixed Income Clearing Corporation, where hedge funds borrow through sponsored repo. Government debt exposure is hedge funds' gross exposure to U.S. government debt, and repo borrowing is the cash they borrow in repo, both from Form PF. Treasury securities include coupons with remaining maturity under thirteen months. Sources: Federal Reserve daily ON RRP series (RRPONTSYD) and SEC Form N-MFP through the OFR Money Market Fund Monitor. Hedge fund rows come from SEC Form PF through the OFR Hedge Fund Monitor, and issuance rows from the Monthly Statement of the Public Debt.

Government/Agency, and Exempt Government funds as government funds. The non-fund remainder is the amount the New York Fed reports for government-sponsored enterprises, banks, and primary dealers. The series starts in 2016:M1. The Federal Reserve ran term reverse repurchase operations at quarter ends in 2014 and 2015, and funds report those alongside the overnight facility while the overnight balance excludes them.

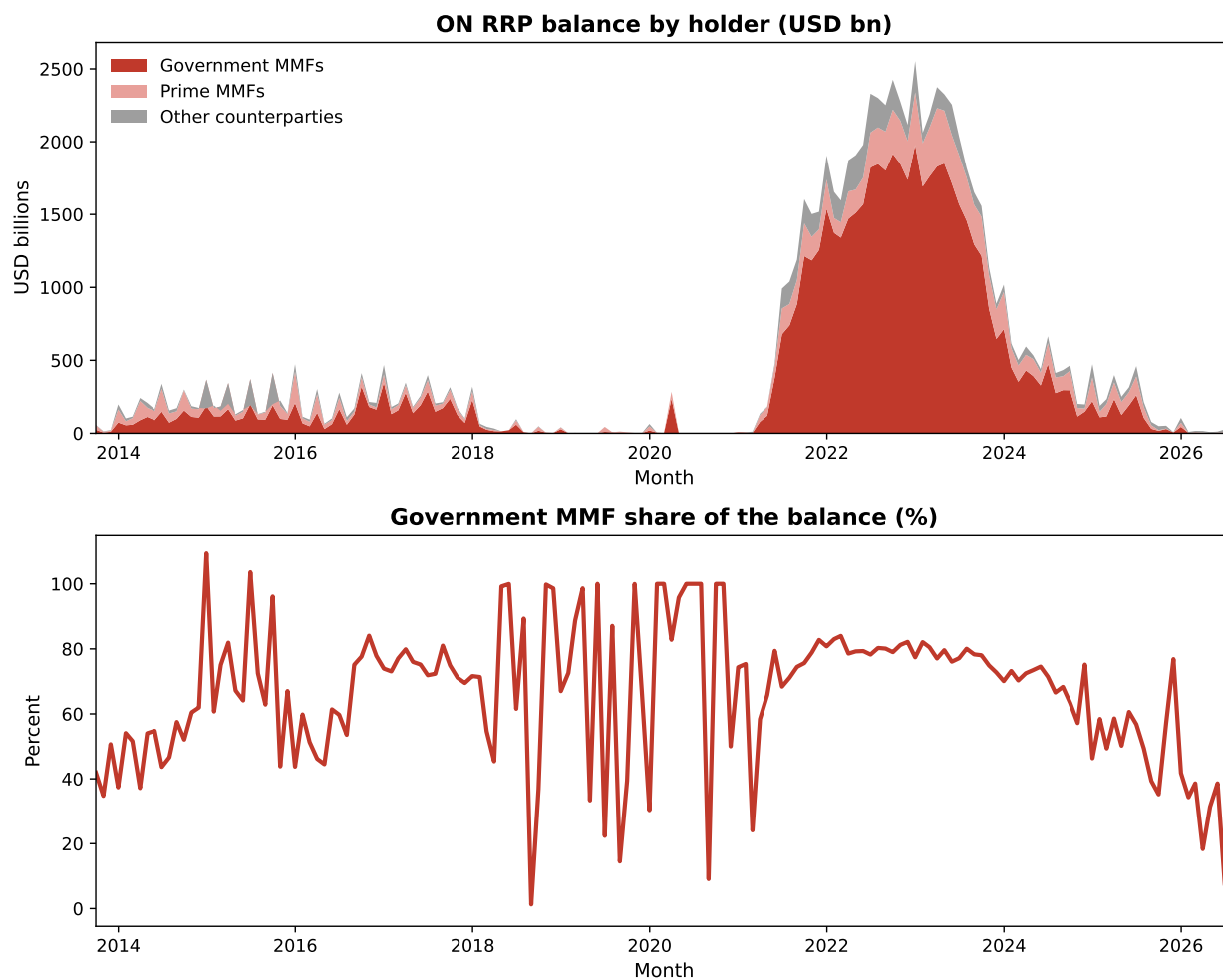


Figure A.1: The ON RRP balance by holder.

Note. Top panel: the month-end ON RRP balance in USD billions, split into government money market funds, prime money market funds, and other counterparties. Bottom panel: the government fund share of the balance in percent. Fund holdings are repurchase agreements with the Federal Reserve Bank of New York reported on Form N-MFP, dated to the last operation of the month; the total and the non-fund remainder are the New York Fed's counterparty-type amounts for that operation. Source: SEC Form N-MFP, Federal Reserve Bank of New York, FRED series RRPONTSYD.

B The Narrative Restrictions

This appendix states the narrative restrictions of [Section 3.3](#). Each restriction ties a labeled shock to the episode that generated it through two requirements. Let W be a window of months inside a regime and j the focal shock. The first is a sign requirement on the shock's realizations. For the RFBS window the mean realization is positive,

$$\frac{1}{|W|} \sum_{t \in W} \tilde{\varepsilon}_{j,t} > 0. \quad (\text{B.1})$$

For the OFBS window the restriction reverses the inequality in (B.1). The second restricts the historical decomposition. Let $\text{HD}_{j \rightarrow L}(W)$ denote the contribution of labeled shock j to $L = \log(\text{Reserves} + \text{ON RRP})$ accumulated over the window W . The restriction requires this contribution to dominate those of the other labeled shocks,

$$|\text{HD}_{j \rightarrow L}(W)| > \sum_{\substack{i \text{ labeled} \\ i \neq j}} |\text{HD}_{i \rightarrow L}(W)|. \quad (\text{B.2})$$

In the baseline, $\text{HD}_{j \rightarrow L}(W)$ is the multi-horizon interval historical decomposition of [Antolín-Díaz and Rubio-Ramírez \(2018\)](#), the contribution of shock j to the level of $\log(\text{Reserves} + \text{ON RRP})$ at the end of the window. The contributions of the n rotated shocks add up to the unexpected movement of $\log(\text{Reserves} + \text{ON RRP})$ over the window, and each restriction assigns a share of that movement to each shock. The sum runs over the labeled shocks only, while [Antolín-Díaz and Rubio-Ramírez \(2018\)](#) compare the focal shock against every other shock.

C HARS and HARS-Z

This appendix expands [Section 4](#). [Appendix C.1](#) reviews HARS in detail. [Appendix C.2](#) derives the zero constraint matrices and states the formal consequences of adding zero restrictions to HARS. [Appendix C.3](#) builds the map γ_z . [Appendix C.4](#) states the results that establish the validity of HARS-Z. [Appendix C.5](#) states the posterior-inference estimator. The next appendix collects the proofs.

C.1 HARS in Detail

The absorbed rotation. HARS (KZ, Algorithm 1) combines likelihood-preserving posterior draws of the model parameters with rotations drawn from a well-defined probability distribution restricted by the sign and narrative restrictions. In homoskedastic SVARs, the rotations of the coefficients, $(A_0, A_+) \mapsto (A_0 Q, A_+ Q)$ with $Q \in O(n)$, are observationally equivalent. Sign restrictions identify shocks by selecting, among these rotations, those that produce economically sensible impulse responses. Under heteroskedasticity this rotation invariance fails. Rotating the coefficients while holding the regime variances fixed mixes shocks whose relative variances differ across regimes and changes the heteroskedastic likelihood (KZ, Lemma 1 and Proposition 2). The rotation would change the fitted model rather than select an interpretation.

KZ's construction absorbs the regime scale before rotation. Define $\Xi_{s_t} := \Lambda_{s_t}^{-1/2}$ and

$$\varepsilon_t := \Xi_{s_t} \eta_t, \quad E(\varepsilon_t \varepsilon_t' \mid s_t) = I_n,$$

and the shocks have unit variance in every regime. Post-multiplying [\(1\)](#) by Ξ_{s_t} absorbs the regime scaling into the block $(A_0 \Xi_{s_t}, A_+ \Xi_{s_t})$. Rotating the entire absorbed block by $Q \in O(n)$ preserves the independent-shock representation and leaves the likelihood unchanged (KZ, Proposition 1),

$$p(y \mid \theta, Q) = p(y \mid \theta) \quad \text{for every } Q \in O(n), \quad (\text{C.1})$$

where $\theta = (A_0, \Lambda, A_+)$ collects the structural objects and $\Lambda = \{\Lambda_k\}_{k=1}^h$. Absorbing first keeps the likelihood invariant to Q under heteroskedasticity. I refer to the rotated unit-variance shocks $\tilde{\varepsilon}'_t = \varepsilon'_t Q$ as the labeled shocks. The dynamic responses of the observables to the labeled shocks in regime k at horizon ℓ are the rows of

$$\Theta_{k,\ell}(Q) = Q' \Lambda_k^{1/2} A_0^{-1} \Phi_\ell(B), \quad (\text{C.2})$$

where $\{\Phi_\ell(B)\}_{\ell \geq 0}$ denotes the moving-average coefficients of the reduced form, with $\Phi_0(B) = I_n$.¹⁹ The regime affects the responses only through the positive diagonal matrix $\Lambda_k^{1/2}$. I also define the average transmission

$$\bar{\Theta}_\ell(Q) = Q' A_0^{-1} \Phi_\ell(B), \quad (\text{C.3})$$

which evaluates the responses at the average of the relative variances across regimes, and that average equals one under the normalization. The average transmission is the object I report as the impulse response. Because Q' mixes the regime-specific scales in $\Lambda_k^{1/2}$ across shocks, the regime-specific responses are not positive rescalings of the average response, and restrictions imposed regime by regime add information.

The restricted posterior target. The likelihood invariance (C.1) separates posterior simulation from identification. Write $SR(\theta, Q)$ for the indicator that the sign and narrative restrictions are satisfied, and define the admissible set $R(\theta) = \{Q \in \mathcal{O}(n) : SR(\theta, Q) > 0\}$, as in KZ. When $\mu(R(\theta)) > 0$, where μ is Haar measure on $\mathcal{O}(n)$, KZ's Definition 1 gives the restricted posterior target

$$\pi(d\theta, dQ \mid y, SR) = p(\theta \mid y) \frac{\mathbf{1}\{Q \in R(\theta)\}}{\mu(R(\theta))} \mu(dQ). \quad (\text{C.4})$$

¹⁹Under the right-multiplication convention, $y'_{t+\ell} = \dots + u'_t \Phi_\ell(B)$ and $\Phi_\ell(B) = \sum_{m=1}^{\min(\ell, p)} \Phi_{\ell-m}(B) B_m$, where B_m is the m -th $n \times n$ lag block of B . Row j of $\Theta_{k,\ell}(Q)$ collects the horizon- ℓ responses of the n observables to labeled shock j . Unlike KZ, who write $\Theta_{k,\ell}(Q) = \Phi_\ell(B) Q' \Lambda_k^{1/2} A_0^{-1}$, this draft places the moving-average matrix on the right, as in (C.2), and all remaining notation is identical.

Drawing θ from its posterior and then drawing Q from Haar measure restricted to $R(\theta)$ produces exact draws from this target (KZ, Proposition 3). The restrictions truncate only the conditional distribution of the rotation. They change the support of Q , not the posterior weight on θ .

The rotation in Gaussian coordinates. HARS draws the rotation in Gaussian coordinates. An $n \times n$ matrix X with independent standard-normal entries maps to a Haar-distributed rotation $Q = \gamma(X)$ through the QR map γ . Column-wise updates of X by the elliptical slice sampler (ESS; Murray et al., 2010) leave the conditional distribution of Q in (C.4) invariant (KZ, Proposition 4). When a posterior move leaves the retained rotation stale, HARS repairs in two stages. The posterior-draw repair redraws completed parameter candidates at the fixed rotation and retains the first admissible one. This repair preserves the sign-restricted marginal target of (A_0, Λ) (KZ, Theorem 1 and Step 2a of their Algorithm 1). Only if no candidate passes does the global rotation repair draw independent rotations from Haar measure and retain the first admissible one. That draw has the restricted-Haar conditional distribution of (C.4) (KZ, Proposition 5(i) and Step 2b).

C.2 The Failure Under Zero Restrictions

Zero restrictions break this construction, and the failure originates in the measure rather than in the algorithm. The zeros invalidate the restricted target (C.4) and the Gaussian coordinatization, while the likelihood invariance (C.1) continues to hold. The obstacle is that a zero restriction is not a truncation. Sign and narrative restrictions truncate the conditional distribution of Q to a subset of positive Haar measure, and the target (C.4) remains well defined. A zero restriction instead imposes a linear equality constraint on a column of Q and lowers the dimension of the admissible set. Adding the zeros to the indicator SR fails, as Remark 1(i) states below.

Under the average transmission (C.3), the impact response of variable i_0 to shock j is $q'_j A_0^{-1} e_{i_0}$, where e_{i_0} is the i_0 -th column of I_n . The response is linear in the j -th column q_j of Q , and a zero restriction on this response is a linear constraint on q_j . For a structural draw θ , I describe the zeros on shock j with two objects, following the notation of Arias et al. (2018). The scalar z_j counts the

zero restrictions on shock j . The $z_j \times n$ matrix $Z_j(\theta)$ stacks those restrictions as rows. Each row is $(A_0^{-1} e_{i_0})'$ for an impact zero on variable i_0 and $(A_0^{-1} \Phi_\ell(B) e_{i_0})'$ for a horizon- ℓ zero, and z_j is the row count of $Z_j(\theta)$.²⁰ The zero restrictions on shock j read

$$Z_j(\theta) q_j = 0,$$

and the sampler must rebuild the constraint matrices at every iteration. The matrices depend on the draw through A_0 and $\Phi_\ell(B)$.²¹ Define the zero manifold

$$\mathcal{Q}(\theta) = \{ Q \in \mathcal{O}(n) : Z_j(\theta) q_j = 0 \text{ for every restricted } j \}.$$

Whenever some $z_j \geq 1$, $\mathcal{Q}(\theta)$ is a lower-dimensional smooth manifold in $\mathcal{O}(n)$ and $\mu(\mathcal{Q}(\theta)) = 0$, as ARRW note in motivating their Algorithm 2.²² The next remark states the consequences for HARS.

Remark 1 (HARS cannot impose zero restrictions). Fix θ with at least one zero restriction. Then $\mu(\mathcal{Q}(\theta)) = 0$, and the following hold.

- (i) The restricted posterior target (C.4) is not defined.
- (ii) The Gaussian coordinatization never reaches the zeros, that is, $\Pr\{\gamma(X) \in \mathcal{Q}(\theta)\} = 0$.

(iii) Every step of KZ's Algorithm 1 fails with probability one. The failure covers the posterior draw

²⁰ARRW impose the zeros on a transformation $F(A_0, A_+)$ of the structural parameters, writing $Z_j F(A_0, A_+) e_j = 0$ with a constant selection matrix Z_j and e_j the j th column of I_n . In their orthogonal reduced-form parameterization the rotation carries the restriction, and their construction evaluates F before the rotation, giving $Z_j F q_j = 0$. I fold that product into a single matrix $Z_j(\theta) := Z_j F(\theta)$, which matches KZ's convention $\tilde{\varepsilon}'_t = \varepsilon'_t Q$. The fold is notational. The transformation F appears as the rows of $Z_j(\theta)$, and the count z_j , the dimension bounds, and the null-space construction are identical to theirs.

²¹Unlike the admissibility check $SR(\theta, Q)$, in which the sign restrictions must hold in every regime and Λ enters, the zero constraints do not depend on Λ . The average transmission sets the relative variances to one, and the zeros hold exactly under (C.3). They need not hold in the regime-specific responses (C.2), where $\Lambda_k^{1/2}$ enters the product $Q' \Lambda_k^{1/2} A_0^{-1} \Phi_\ell(B)$ and reweights the constrained column.

²²When $Z_j(\theta)$ has full row rank, the column q_j must lie on the unit sphere of the $(n - z_j)$ -dimensional null space of $Z_j(\theta)$. The zero restrictions lower the dimension of $\mathcal{Q}(\theta)$ below the $n(n - 1)/2$ dimensions of $\mathcal{O}(n)$, and any lower-dimensional subset has Haar measure zero. ARRW establish the same fact in the structural parameterization. The set of structural parameters satisfying the sign and zero restrictions is of Lebesgue measure zero in the set of all structural parameters. The zero restrictions define a lower-dimensional smooth manifold, and the volume measure of their Theorem 3 serves as the reference measure on it.

with admissibility check (Step 1), the posterior-draw repair by the joint procedure (Step 2a), the global rotation repair (Step 2b), and the rotation draw (Step 3). The column-by-column accept–reject initialization fails as well.

Proof of Remark 1. (i) Adding the zeros to the indicator SR forces $R(\theta) \subseteq Q(\theta)$. Monotonicity gives $\mu(R(\theta)) \leq \mu(Q(\theta)) = 0$. The condition $\mu(R(\theta)) > 0$ of KZ’s Definition 1 fails at every θ with at least one zero restriction.

(ii) The map γ is the QR map that normalizes the R factor to a positive diagonal and returns the orthogonal factor. The rotation $\gamma(X)$ is Haar distributed on $O(n)$ (KZ, Proposition 4). The equality $\mu(Q(\theta)) = 0$ gives $\Pr\{\gamma(X) \in Q(\theta)\} = 0$. The construction projects each Gaussian column only onto the orthogonal complement of the preceding columns, never onto the null spaces of $Z_j(\theta)$.

(iii) The rotation-side steps of KZ’s Algorithm 1 fail by (ii). Only Steps 1 and 2a need a separate argument. Both redraw θ rather than Q .

Initialization. Each accepted column of the initialization, Algorithm 4 of [Arias et al. \(2026\)](#), is a projected Gaussian draw. A projected Gaussian draw lies in the null space of $Z_j(\theta)$ with probability zero, by (ii). Almost surely no initial admissible pair exists. The chain never occupies a state whose rotation satisfies the zeros.

Steps 1 and 2a. Both hold the retained rotation fixed and check completed draws. For a fixed unit column $q_j \neq 0$, the condition $Z_j(\theta) q_j = 0$ depends on θ only through A_0 and $\Phi_\ell(B)$. The condition is an exact alignment among functions of A_0^{-1} and $\Phi_\ell(B)$. The set of structural parameters at which the alignment holds is of Lebesgue measure zero, following the argument ARRW use for their set U (their Appendix A.3). Every completed draw and candidate comes from a proposal density and the Gibbs conditional density $p_+(A_+ \mid z_0, s_0)$. Draws with a density avoid the alignment set almost surely. When the MH step rejects and sets $z_0^\star = z_0$, the zeros already fail at the current first block, by the initialization argument.

Step 2b. Each independent Haar draw lies off $Q(\theta^\star)$ almost surely, by (ii).

Step 3. The slice step starts from a current admissible point. No such point is ever available, by the initialization argument. Even from a state on $Q(\theta)$, the update fails. Consider the ESS update of

a column j that carries zero restrictions. Following KZ's Proposition 4, draw an auxiliary vector $\tau_j \sim \mathcal{N}(0, I_n)$ and consider the ellipse

$$x_j(\vartheta) = x_j \cos \vartheta + \tau_j \sin \vartheta, \quad \vartheta \in [0, 2\pi].$$

Under γ , the proposed column is the projection of $x_j(\vartheta)$ onto the orthogonal complement of the preceding columns, normalized. Write P for that projection. The projection P is fixed along the ellipse. The preceding columns do not move when the ESS step updates column j . The normalization does not affect the zero condition. The zero restrictions hold at the proposal if and only if $Z_j(\theta) P x_j(\vartheta) = 0$. Linearity of P gives

$$Z_j(\theta) P x_j(\vartheta) = \cos \vartheta Z_j(\theta) P x_j + \sin \vartheta Z_j(\theta) P \tau_j.$$

The coefficient $Z_j(\theta) P x_j$ is zero at the current state. The current column satisfies its zeros and is proportional to $P x_j$. The coefficient $Z_j(\theta) P \tau_j$ is nonzero almost surely. The auxiliary vector is a fresh Gaussian draw, and $P \tau_j$ is a projected Gaussian column. A projected Gaussian column satisfies its zero restrictions with probability zero, by (ii). The zero restrictions hold on the ellipse only where $\sin \vartheta = 0$, at $\vartheta \in \{0, \pi\}$. At these angles the proposal is the current column up to sign. The update never reaches a new rotation on $\mathcal{Q}(\theta)$. The slice step retains only angles at which the admissibility indicator passes, and no angle off $\{0, \pi\}$ passes. $\square$

Rotations drawn from the uniform distribution satisfy the zero restrictions with probability zero.²³ The sampler must parameterize the zero-restricted set directly rather than reach it by sampling. It must build the rotation on $\mathcal{Q}(\theta)$, and its slice-sampling updates must never leave it.

Appendix C.3 constructs exactly such a parameterization.

²³The obstruction is also topological. The elliptical slice sampler of [Arias et al. \(2026\)](#) assumes an admissible set of positive measure (their Assumption 1). Strict-inequality sign restrictions give an open set of positive measure. Zero restrictions give a set that is neither open nor of positive measure.

C.3 Extending HARS to Zero Restrictions

HARS-Z extends HARS to include zero restrictions on the rotation. Developing this extension is the paper's methodological contribution. The extension replaces the QR map γ with a map $\gamma_z(\cdot; \theta)$ from a Gaussian matrix to a rotation that lies on $Q(\theta)$ *by construction*. ARRW's Algorithm 2 supplies such a map, and the zero restrictions hold exactly along the chain. The subsection builds the map γ_z and sets the state of the chain and its posterior target. It identifies the measure that the target induces on $Q(\theta)$ and constructs the rotation draw.

ARRW's Algorithm 2 builds the columns of Q one at a time, following a permutation σ of the shock indices $\{1, \dots, n\}$ that places the most-restricted shocks first. The index $\sigma(j)$ is the shock whose column the construction builds at step j . The permutation must satisfy $z_{\sigma(j)} \leq n - j$ for every j , where $z_{\sigma(j)}$ is the zero count of the shock assigned to step j . The j -th column must be orthogonal to the $j - 1$ preceding columns and must satisfy its own $z_{\sigma(j)}$ zeros. A candidate direction exists only if $(j - 1) + z_{\sigma(j)} < n$.²⁴ For every restricted shock j , the matrix $Z_j(\theta)$ must also have full row rank, that is, its z_j rows must be non-redundant.²⁵

Given a Gaussian matrix $G = [g_1, \dots, g_n]$, the construction proceeds one column at a time. At step j , the admissible directions for column $q_{\sigma(j)}$ form the linear subspace

$$N_j(\theta) := \text{null}(M_j(\theta)), \quad M_j(\theta) := \begin{bmatrix} [q_{\sigma(1)}, \dots, q_{\sigma(j-1)}]' \\ Z_{\sigma(j)}(\theta) \end{bmatrix}, \quad (\text{C.5})$$

where $M_j(\theta)$ stacks the preceding columns $q_{\sigma(1)}, \dots, q_{\sigma(j-1)}$ and the zero restrictions of shock $\sigma(j)$. The null space $N_j(\theta)$ collects the vectors orthogonal to every preceding column and consistent with those zeros.

The projection at step j requires the rows of $M_j(\theta)$ to be linearly independent. The null space has the dimension $n - (j - 1) - z_{\sigma(j)}$ that the ordering condition above presumes. Each block of $M_j(\theta)$ is independent on its own. The preceding columns are orthonormal, and $Z_{\sigma(j)}(\theta)$ has full row rank.

²⁴Later positions have more orthogonality constraints and admit fewer zeros, and the most heavily restricted shocks come first.

²⁵ARRW impose this assumption on their selection matrices Z_j when they set their notation (their Section 2.2).

The requirement binds only across the two blocks. I assume that the rows of $Z_{\sigma(j)}(\theta)$ are linearly independent of the preceding columns at every orthonormal collection $q_{\sigma(1)}, \dots, q_{\sigma(j-1)}$ whose columns satisfy their zero restrictions.²⁶ ARRW derive this property from a regularity condition on their function F and the full row rank of the Z_j . See their footnote 6 in Section 2.2 and their Appendix A.3, conditions 1 and 2. I assume it directly, at each θ the chain visits rather than over the whole parameter space.

The construction projects g_j orthogonally onto $N_j(\theta)$ and normalizes,

$$q_{\sigma(j)} = \frac{P_j(\theta) g_j}{\|P_j(\theta) g_j\|}, \quad (\text{C.6})$$

where $P_j(\theta)$ denotes the orthogonal projection onto $N_j(\theta)$ and the denominator is nonzero for almost every g_j . Write $\gamma_z(\cdot; \theta)$ for the map $G \mapsto Q$ that these n projection steps define. The chain below stores this Gaussian input as part of its state and denotes it by X .

Two properties follow. First, each column satisfies its constraints by construction, and after n steps Q is orthogonal and lies on $Q(\theta)$. The construction involves no accept–reject step. Second, the projected Gaussian column favors no direction within $N_j(\theta)$, and the normalized column is uniform over the admissible directions at step j .²⁷ This step-by-step distribution matches Steps 2 and 3 of ARRW’s Algorithm 2.²⁸

The uniform distribution of each column, conditional on the preceding columns, determines the law of the completed rotation. Write $\mu_z(\cdot; \theta)$ for the null-space construction measure on $Q(\theta)$, the

²⁶This independence fails only at draws of θ where some combination of the rows of $Z_{\sigma(j)}(\theta)$ falls exactly in the span of the preceding columns. The rows are functions of entries of A_0^{-1} and the moving-average coefficients. The failure requires an exact equality among these entries. A draw of θ from a continuous distribution satisfies an exact equality with probability zero. The retained draws avoid the failure set almost surely. ARRW make the same genericity argument for their set U (their Appendix A.3), and the implementation checks the condition at every construction step at no additional cost.

²⁷The orthogonal projection of a standard Gaussian vector onto a linear subspace is a standard Gaussian within that subspace, whose distribution is invariant under rotations of the subspace. Its normalized direction is uniform on the subspace’s unit sphere.

²⁸At each step j , they draw a standard normal vector on $\mathbb{R}^{n+1-j-z_j}$, normalize it to the unit sphere, and map it through a matrix K_j . The columns of K_j form an orthonormal basis for the null space of M_j , the matrix that stacks the preceding columns and the zero restrictions. The orthogonal projection above induces the same uniform distribution on that null space. The two constructions draw each column from the same distribution.

distribution of $\gamma_z(G; \theta)$ under a standard Gaussian matrix G ,

$$\mu_z(C; \theta) := \Pr\{\gamma_z(G; \theta) \in C\} \quad \text{for every measurable } C \subseteq \mathcal{Q}(\theta). \quad (\text{C.7})$$

Proposition 1 characterizes this measure and relates it to the conditional-Haar (uniform) measure on $\mathcal{Q}(\theta)$.

I depart from ARRW in one respect. At step j they draw $x_j \in \mathbb{R}^{n+1-j-z_j}$ from a standard normal distribution and normalize it to the unit sphere. The dimension of their coordinates varies with j and, through z_j , with the restriction pattern. I instead use a single $n \times n$ Gaussian matrix X , the coordinates in which HARS runs its ESS update. Step j takes the j -th column of X as its input. The randomness remains separate across steps, and every column keeps the fixed dimension n . The projection at each step keeps only the components in $N_j(\theta)$, and the projected column has the same distribution as the vector ARRW draw at that step. The fixed $n \times n$ matrix allows me to apply the column-wise ESS update of KZ (Proposition 4) and its validity argument without change.

Lemma 1 (Properties of the map γ_z). *Fix θ such that, for every j , the matrix $Z_{\sigma(j)}(\theta)$ has full row rank and the permutation satisfies $z_{\sigma(j)} \leq n - j$. Require also that the matrix $M_j(\theta)$ in (C.5) has full row rank at every orthonormal collection $q_{\sigma(1)}, \dots, q_{\sigma(j-1)}$ whose columns satisfy their zero restrictions. Then the following hold for almost every G .*

- (i) $\gamma_z(G; \theta) \in \mathcal{Q}(\theta)$, and every zero restriction holds exactly, with no rejection.
- (ii) Every construction step k depends on G only through $g_1, \dots, g_k$. Perturbing g_j leaves the columns from steps $1, \dots, j - 1$ unchanged, and the construction re-projects the perturbed and later columns into their own null spaces.
- (iii) If $Q \in \mathcal{Q}(\theta)$ is admissible and $G_Q := [q_{\sigma(1)}, \dots, q_{\sigma(n)}]$ collects the columns of Q in build order, then $\gamma_z(G_Q; \theta) = Q$. The construction reproduces its own output, and setting $X = G_Q$ places the chain at an admissible state.

Proof of Lemma 1. (i) At step j the subspace $N_j(\theta)$ in (C.5) is, by construction, orthogonal to

every preceding column and contained in $\text{null}(Z_{\sigma(j)}(\theta))$. The projected and normalized column $q_{\sigma(j)}$ therefore has unit norm, is orthogonal to its predecessors, and satisfies its zero restrictions exactly. The matrix $M_j(\theta)$ in (C.5) has $(j - 1) + z_{\sigma(j)}$ rows. The dimension of $N_j(\theta)$ equals $n + 1 - j - z_{\sigma(j)} \geq 1$ if and only if $M_j(\theta)$ is of full row rank (ARRW). Full row rank holds by hypothesis at every orthonormal collection of preceding columns whose columns satisfy their zero restrictions. The construction completes with Q orthogonal. The projection of a Gaussian column is zero only on a null set. The rotation $\gamma_z(G; \theta)$ lies in $Q(\theta)$ for almost every G .

(ii) Fix a step k . The subspace $N_k(\theta)$ has two ingredients, the preceding columns and the constraint matrix $Z_{\sigma(k)}(\theta)$. The constraint matrix is fixed at the given θ . The preceding columns come from steps $1, \dots, k - 1$, and step k adds g_k and nothing else. By induction, every construction step k depends on G only through $g_1, \dots, g_k$. In particular, g_j is not among the inputs of steps $1, \dots, j - 1$ and is among the inputs of steps $j, \dots, n$. Perturbing g_j leaves steps $1, \dots, j - 1$ untouched. The construction recomputes steps $j, \dots, n$. Each recomputed column passes through the projection onto its own null space. The zeros hold after the perturbation.

(iii) Let $Q \in Q(\theta)$ be admissible. Each of its columns already lies in its own subspace $N_j(\theta)$. Orthogonal projection acts as the identity on $N_j(\theta)$. Normalization preserves a unit vector. The construction reproduces Q column by column. The reproduction property is the projection argument ARRW use to establish that their Algorithm 2 can generate every rotation satisfying the zero restrictions. $\square$

Lemma 1 makes ARRW's Algorithm 2 usable inside a Markov chain. Part (i) restates the property of their Algorithm 2 in the fixed $n \times n$ Gaussian parameterization introduced above. Part (ii) is new. ARRW draw each rotation independently and never perturb one afterward. They never need to ask how the rotation changes when a single Gaussian column moves.²⁹ The column-wise ESS update moves one column of X along an ellipse and poses exactly that question. Part (ii) supplies the answer. The recomputed rotation stays on $Q(\theta)$. Part (iii) reuses ARRW's support argument

²⁹ARRW use their coordinates only for independent proposals. The QR map γ of KZ fails for the reason given in Remark 1(ii). Under γ , an elliptical perturbation of a single Gaussian column cannot restore the zeros, and the sign-only ESS never visits the manifold.

for a new purpose. Setting $X = G_Q$ starts the chain at any admissible rotation, whatever search produced that rotation. Part (iii) supplies the admissible state that [Algorithm 1](#) presupposes.

The state of the chain. A sampler could store the rotation itself, keeping the matrix Q in the state and reusing it after every structural update. HARS does exactly this. KZ write the state as (θ, Q) and set $X = Q$ whenever the ESS step needs Gaussian coordinates. The choice is harmless there. The QR map γ does not depend on θ , and the stored rotation and its coordinates never disagree. On $Q(\theta)$ the choice matters. The manifold $Q(\theta)$ moves with θ . A stored Q lies on the old manifold and leaves the new manifold $Q(\theta^*)$ with probability one as soon as the structural draw moves. The zero restrictions fail after every structural update. I store the Gaussian coordinates instead. The state is the pair (θ, X) , where X is an $n \times n$ Gaussian matrix, and the sampler recomputes the rotation from the state whenever it needs one,

$$Q = \gamma_z(X; \theta).$$

The next subsection establishes the validity of the construction. [Proposition 1](#) derives the measure that the construction induces on the admissible set, [Corollary 2](#) shows that the weighted draws recover conditional-Haar expectations, and [Proposition 2](#) gives the column-wise elliptical slice sampling update that the sampler uses. [Algorithm 1](#) collects the steps.

C.4 The Validity of HARS-Z

A structural update leaves X unchanged and changes only the rotation $\gamma_z(X; \theta)$. The construction projects the same Gaussian columns onto the null spaces of the new draw, and $\gamma_z(X; \theta^*)$ lies on $Q(\theta^*)$ by [Lemma 1\(i\)](#). The zeros hold after every structural move by construction, and the admissibility check reduces to the sign and narrative restrictions. The rotation remains auxiliary, exactly as in KZ. Given (θ, X) , the rotated absorbed system of [\(C.1\)](#) evaluated at $Q = \gamma_z(X; \theta)$ generates the data, and the conditional density $p(y \mid \theta, X)$ is well defined. The next definition states the posterior target of HARS-Z, and the corollary that follows establishes its validity.

Definition 1 (The restricted posterior target on (θ, X)). Let $\varphi(X)$ denote the standard Gaussian density of the $n \times n$ matrix X and define

$$\kappa(\theta) = \int \mathbf{1}\{SR(\theta, \gamma_z(X; \theta)) > 0\} \varphi(X) dX,$$

the Gaussian probability of the admissible set at θ . When $\kappa(\theta) > 0$, define the conditional distribution of X given θ ,

$$\pi_X(dX \mid \theta, SR) := \frac{\mathbf{1}\{SR(\theta, \gamma_z(X; \theta)) > 0\} \varphi(X)}{\kappa(\theta)} dX. \quad (\text{C.8})$$

The restricted posterior target is

$$\pi(d\theta, dX \mid y, SR) = \pi(\theta \mid y) \pi_X(dX \mid \theta, SR) d\theta. \quad (\text{C.9})$$

[Definition 1](#) is KZ's Definition 1 in Gaussian coordinates, with $\kappa(\theta) > 0$ replacing $\mu(R(\theta)) > 0$. KZ's definition takes one fixed distribution, Haar measure on $O(n)$, and restricts it to the θ -dependent region $R(\theta)$. That structure is unavailable on rotations here. A distribution on $Q(\theta)$ assigns no mass to $Q(\theta')$ at any other draw θ' , and no fixed distribution on rotations exists. The Gaussian distribution of X restores the structure. Equation (C.8) restricts the same $\varphi(X)$ to the admissible set at every draw, and θ enters only through the indicator and the map $\gamma_z(\cdot; \theta)$.

Corollary 1 (The Gaussian coordinates are auxiliary). *Fix θ satisfying the conditions of [Lemma 1](#) and write $Q = \gamma_z(X; \theta)$. For almost every X ,*

$$p(y \mid \theta, X) = p(y \mid \theta, Q) = p(y \mid \theta).$$

If, in addition, $\kappa(\theta) > 0$, then drawing θ from its posterior and then drawing X from the conditional $\pi_X(\cdot \mid \theta, SR)$ in (C.8) produces exact draws from the joint target (C.9). Sampling X in a separate block leaves the marginal posterior of θ unchanged.

Proof of [Corollary 1](#). Fix θ and the regime path. Given (θ, X) , the rotated absorbed system

evaluated at $Q = \gamma_z(X; \theta)$ generates the data,

$$y_t' A_0 \Xi_{s_t} Q = x_t' A_+ \Xi_{s_t} Q + \tilde{\varepsilon}_t', \quad E(\tilde{\varepsilon}_t \tilde{\varepsilon}_t' | s_t) = I_n,$$

where $\tilde{\varepsilon}_t' = \varepsilon_t' Q = \eta_t' \Xi_{s_t} Q$ collects the labeled shocks of Appendix C.1. The conditional density of y given (θ, X) is $p(y | \theta, Q)$ evaluated at $Q = \gamma_z(X; \theta)$. The evaluation gives the first equality of the corollary. By Lemma 1(i), $\gamma_z(X; \theta) \in \mathcal{Q}(\theta) \subseteq \mathcal{O}(n)$ for almost every X . KZ's Proposition 1 gives $p(y | \theta, Q) = p(y | \theta)$ for every $Q \in \mathcal{O}(n)$. Evaluating at $Q = \gamma_z(X; \theta)$ yields the second equality of the corollary. The dependence of the map on θ does not affect the evaluation. The invariance $p(y | \theta, Q) = p(y | \theta)$ holds at each fixed θ for every $Q \in \mathcal{O}(n)$, and $\gamma_z(X; \theta)$ is one such rotation.

For the last claim, let $\kappa(\theta) > 0$. The conditional $\pi_X(\cdot | \theta, SR)$ of (C.8) is then well defined. The conditional is the Gaussian distribution of X restricted to the admissible set $\{X : SR(\theta, \gamma_z(X; \theta)) > 0\}$. Specify the prior of the pair as $p(\theta)$ for θ and $\pi_X(\cdot | \theta, SR)$ for X given θ . The joint posterior under this prior satisfies

$$\begin{aligned} \pi(\theta, X | y) &\propto p(y | \theta, X) p(\theta) \pi_X(X | \theta, SR) \\ &= p(y | \theta) p(\theta) \pi_X(X | \theta, SR) \\ &\propto \pi(\theta | y) \pi_X(X | \theta, SR). \end{aligned}$$

The first line is Bayes' rule. The second line applies the first part. The third line collects $\pi(\theta | y) \propto p(y | \theta) p(\theta)$. The conditional distribution of X is the Gaussian distribution restricted to the admissible set. The posterior density of θ remains $\pi(\theta | y)$. The joint posterior under this prior is exactly the target (C.9). Drawing θ from its posterior and then drawing X from $\pi_X(\cdot | \theta, SR)$ produces exact draws from the target. The argument is KZ's Proposition 3 with Haar measure restricted to $R(\theta)$ replaced by the Gaussian distribution of X restricted to the admissible set. $\square$

Corollary 1 justifies the two-block structure of the sampler. The equalities say that the likelihood never involves X . The θ block runs the posterior sampler of BPSS exactly as in KZ. The last claim

covers the rotation block. Drawing the rotation at fixed θ from the conditional (C.8) produces draws from the target (C.9) and leaves the marginal posterior of θ unchanged.

Posterior inference evaluates the impulse responses, functions of (θ, Q) . The next definition states the distribution that posterior inference targets.

Definition 2 (The conditional-Haar target on (θ, Q)). Fix the conditions of Lemma 1 at every θ with $\kappa(\theta) > 0$. The conditional-Haar target, restated from Section 4, is

$$\pi_v(d\theta, dQ \mid y, SR) = \pi(\theta \mid y) \frac{\mathbf{1}\{Q \in R(\theta)\} \mu(dQ \mid Q(\theta))}{\mu(R(\theta) \mid Q(\theta))} d\theta, \quad (6)$$

where $\mu(\cdot \mid Q(\theta))$ denotes the conditional-Haar (uniform) measure on $Q(\theta)$ and $R(\theta) = \{Q \in Q(\theta) : SR(\theta, Q) > 0\}$ denotes the admissible set on the zero-restricted set.

Remark 2. The normalizer of (6) is positive whenever $\kappa(\theta) > 0$. The admissible set $R(\theta)$ is relatively open in $Q(\theta)$, as the restrictions hold with strict inequalities. When $R(\theta)$ is empty, no draw satisfies the indicator in (C.8) and $\kappa(\theta) = 0$. The uniform measure on $Q(\theta)$ assigns positive mass to every nonempty relatively open subset.

Definition 2 is the analogue of KZ's target (C.4), with Haar measure on $\mathcal{O}(n)$ replaced by the conditional-Haar measure on $Q(\theta)$. The restrictions truncate only the conditional distribution of the rotation, as in Proposition 3 of KZ, and the marginal posterior of θ remains $\pi(\theta \mid y)$, consistent with Corollary 1. Definition 1 states the target on (θ, X) , the coordinates in which the chain runs, and Definition 2 states the target on (θ, Q) , the coordinates of the impulse responses.

Proposition 1 (The induced measure on $Q(\theta)$). Fix θ satisfying the conditions of Lemma 1 with $\kappa(\theta) > 0$. The distribution over $Q(\theta)$ that $\pi_X(\cdot \mid \theta, SR)$ induces through γ_z is the null-space construction measure $\mu_z(\cdot; \theta)$ of (C.7), restricted to the admissible set $R(\theta)$. The conditional of (6) given θ is the conditional-Haar measure restricted to $R(\theta)$, written $\pi_v(dQ \mid \theta, y, SR)$. The restricted construction measure has two properties.

- (a) When the zeros restrict a single column of Q and the ordering builds that column first, the restricted construction measure coincides with $\pi_v(\cdot \mid \theta, y, SR)$.

(b) Otherwise the restricted construction measure differs from $\pi_v(\cdot \mid \theta, y, SR)$ by the volume element $v(Q; \theta)$ of the construction (ARRW, Theorem 3), and weighting each retained draw by $v(Q; \theta)$ recovers $\pi_v(\cdot \mid \theta, y, SR)$.

Proof of Proposition 1. Two preliminary steps characterize the induced distribution. Parts (a) and (b) then follow in turn.

Step 1: π_X induces the construction measure restricted to the admissible set. The argument follows the image claim of KZ's Proposition 4, with the QR map γ replaced by γ_z and Haar measure replaced by the construction measure $\mu_z(\cdot; \theta)$. Recall the conditional (C.8),

$$\pi_X(dX \mid \theta, SR) = \frac{\mathbf{1}\{SR(\theta, \gamma_z(X; \theta)) > 0\} \varphi(X)}{\kappa(\theta)} dX.$$

Without the indicator the conditional reduces to the Gaussian density φ . The matrix X then has the same distribution as the Gaussian matrix G in (C.7). The relation $\gamma_z(X; \theta) \sim \mu_z(\cdot; \theta)$ holds by definition.

Now restore the indicator. The indicator in (C.8) does not test X directly. The indicator builds the rotation $\gamma_z(X; \theta)$ and tests whether that rotation satisfies the sign and narrative restrictions. The input X passes exactly when its rotation lies in the admissible set $R(\theta) = \{Q \in O(n) : SR(\theta, Q) > 0\}$ of Section 4,

$$\{X : SR(\theta, \gamma_z(X; \theta)) > 0\} = \{X : \gamma_z(X; \theta) \in R(\theta)\}. \quad (\text{C.10})$$

Take any set C of rotations. Under the conditional $\pi_X(\cdot \mid \theta, SR)$, the probability that the rotation lands in C is

$$\begin{aligned} \Pr\{\gamma_z(X; \theta) \in C \mid SR(\theta, \gamma_z(X; \theta)) > 0\} &= \Pr\{\gamma_z(X; \theta) \in C \mid \gamma_z(X; \theta) \in R(\theta)\} \\ &= \frac{\Pr\{\gamma_z(X; \theta) \in C \cap R(\theta)\}}{\Pr\{\gamma_z(X; \theta) \in R(\theta)\}} \\ &= \frac{\mu_z(C \cap R(\theta); \theta)}{\mu_z(R(\theta); \theta)}. \end{aligned}$$

The first line rewrites the conditioning event. Equation (C.10) shows the two events are the same set of X . The second line applies the definition of conditional probability. The probabilities on the second line are unconditional. On the second line $X \sim \varphi$, and the relation $\gamma_z(X; \theta) \sim \mu_z(\cdot; \theta)$ from the start of the proof evaluates each probability as a μ_z -value on the third line.

The set C is arbitrary, and the values on every set determine a distribution. The induced distribution is

$$\frac{\mathbf{1}\{SR(\theta, Q) > 0\} \mu_z(dQ; \theta)}{\mu_z(R(\theta); \theta)}. \quad (\text{C.11})$$

The measure (C.11) is the construction measure restricted to the admissible set $R(\theta)$. The restricted construction measure is the counterpart of KZ's Haar measure restricted to $R(\theta)$.

The normalizer of (C.11) coincides with the normalizer $\kappa(\theta)$ of (C.8), $\kappa(\theta) = \mu_z(R(\theta); \theta)$. Both equal the probability of one event, that the rotation is admissible. The normalizer $\kappa(\theta)$ computes this probability in X . The normalizer $\mu_z(R(\theta); \theta)$ computes it on $Q(\theta)$.

Step 2: relation to conditional-Haar. The comparison proceeds column by column. The construction builds the rotation column by column. The construction generates $q_{\sigma(1)}, \dots, q_{\sigma(n)}$ in the order σ . Each step projects a fresh Gaussian column onto the null space $N_j(\theta)$ that the zero restrictions and the previously built columns determine, then normalizes. The projected Gaussian column favors no direction within $N_j(\theta)$. Conditional on the preceding columns,

$$q_{\sigma(j)} \mid q_{\sigma(1)}, \dots, q_{\sigma(j-1)} \sim \text{uniform on the unit sphere of } N_j(\theta).$$

The uniform law of the projected column is the second property of the construction in Appendix C.3. Write $\mu_j(\cdot \mid q_{\sigma(1)}, \dots, q_{\sigma(j-1)}; \theta)$ for this conditional distribution, the uniform distribution on the unit sphere of $N_j(\theta)$. The chain rule of probability writes the joint law as the product of these conditionals,

$$\mu_z(dQ; \theta) = \prod_{j=1}^n \mu_j(dq_{\sigma(j)} \mid q_{\sigma(1)}, \dots, q_{\sigma(j-1)}; \theta). \quad (\text{C.12})$$

Equation (C.7) defines $\mu_z(\cdot; \theta)$ only as the distribution of $\gamma_z(G; \theta)$. The product (C.12) makes the

measure explicit. Parts (a) and (b) below use (C.12).

Part (a): the zeros load on a single column. The ordering builds the restricted column first. The first null space $N_1(\theta)$ then equals the null space of the zero restrictions. Every later null space $N_j(\theta)$, $j \geq 2$, is the orthogonal complement of the preceding columns and carries no further constraint.

The comparison involves two distinct measures on $\mathcal{Q}(\theta)$. The construction measure $\mu_z(\cdot; \theta)$ is the law the sampler induces. The measure $\mu(\cdot \mid \mathcal{Q}(\theta))$ is the conditional-Haar (uniform) measure on $\mathcal{Q}(\theta)$ and serves as the reference measure of the target. The subscript z denotes the measure that the zero-restricted construction induces. The conditioning on $\mathcal{Q}(\theta)$ denotes Haar measure restricted to $\mathcal{Q}(\theta)$. Factor both measures at the first column,

$$\mu_z(dQ; \theta) = \mu_1(dq_{\sigma(1)}; \theta) \mu_z(dq_{\sigma(2)}, \dots, dq_{\sigma(n)} \mid q_{\sigma(1)}; \theta), \quad (\text{C.13})$$

$$\mu(dQ \mid \mathcal{Q}(\theta)) = \mu(dq_{\sigma(1)} \mid \mathcal{Q}(\theta)) \mu(dq_{\sigma(2)}, \dots, dq_{\sigma(n)} \mid q_{\sigma(1)}, \mathcal{Q}(\theta)), \quad (\text{C.14})$$

and compare factor by factor.

The first factors of (C.13) and (C.14) coincide. The first factor of (C.13) is $\mu_1(\cdot; \theta)$, the uniform distribution on the unit sphere of $N_1(\theta)$, by the second property of the construction. The first factor of (C.14) is uniform on the same sphere, by the following symmetry argument. Take any orthogonal matrix R that rotates $N_1(\theta)$ within itself and fixes its orthogonal complement. The map $Q \mapsto RQ$ moves the first column along the sphere and leaves every zero restriction intact. The map $Q \mapsto RQ$ therefore sends $\mathcal{Q}(\theta)$ onto itself. Haar measure is invariant under this map. Conditional-Haar inherits the invariance. Every rotation of $N_1(\theta)$ leaves the marginal distribution of $q_{\sigma(1)}$ unchanged. The only probability distribution on a sphere that every rotation leaves unchanged is the uniform distribution. Both first factors are the same uniform law on the same sphere.

The second factors of (C.13) and (C.14) coincide as well. Fix $q_{\sigma(1)}$. No zero restriction remains after the first column. Every later null space is the orthogonal complement of the preceding columns inside $q_{\sigma(1)}^\perp$. Each remaining step projects a fresh Gaussian column onto that complement and normalizes. A projected standard Gaussian column is standard Gaussian within the complement.

The remaining steps therefore run the QR map γ of KZ on $q_{\sigma(1)}^\perp$. The rotation $\gamma(X)$ of a Gaussian matrix is Haar distributed, by the Gaussian-to-Haar construction of [Rubio-Ramírez et al. \(2010\)](#) in KZ’s Proposition 4. The second factor of (C.13) is the uniform distribution over the orthonormal completions of $q_{\sigma(1)}$.

The second factor of (C.14) is the same uniform distribution. Apply the symmetry argument of the first factor to rotations that fix $q_{\sigma(1)}$ and rotate its complement. Every such rotation preserves $Q(\theta)$ and preserves conditional-Haar. The conditional law of the completions is rotation invariant. The uniform distribution is the only rotation-invariant law. ARRW report constant importance weights for their Algorithm 3 without zero restrictions, consistent with the equality of the two factors.

The factorizations (C.13) and (C.14) coincide factor by factor. The two measures are equal,

$$\mu_z(\cdot; \theta) = \mu(\cdot \mid Q(\theta)).$$

The restrictions of equal measures to $R(\theta)$ are equal. The volume element is constant. This proves part (a).

Part (b): the zeros load on two or more columns. The construction still draws each column uniformly on the unit sphere of its null space, as in (C.12). The measure $\mu_z(\cdot; \theta)$ is the law of the construction that ARRW’s Theorem 3 describes. Their change-of-variable theorem for maps into lower-dimensional smooth manifolds gives $\mu_z(\cdot; \theta)$ a density proportional to $1/v(Q; \theta)$ with respect to the uniform (volume) measure on $Q(\theta)$. The factor $v(Q; \theta)$ is the volume element of the map implied by their Algorithm 2 at fixed θ . The ratio of the two densities gives

$$\frac{d\mu(\cdot \mid Q(\theta))}{d\mu_z(\cdot; \theta)}(Q) \propto v(Q; \theta).$$

The density relation holds on all of $Q(\theta)$ and hence on $R(\theta)$. Weighting each retained draw by $v(Q; \theta)$ recovers the conditional-Haar measure restricted to $R(\theta)$, the conditional $\pi_v(\cdot \mid \theta, y, SR)$ of (6). The weighting is necessary. As ARRW document, the distribution that their Algorithm 2

implies is not invariant to a reordering of the shocks. This proves part (b) and the proposition. $\square$

Proposition 1 states that the conditional $\pi_X(\cdot \mid \theta, SR)$ induces the construction measure on $Q(\theta)$ through γ_z , not the uniform (conditional-Haar) measure on $Q(\theta)$. When the zeros load on a single column and the ordering builds that column first, the two coincide and need no correction. With zeros on more than one column, I weight the retained rotations by the volume element and normalize the weights within each retained draw of θ .³⁰ **Proposition 1**(b) states the correction as a density, known only up to a constant that depends on θ . The next corollary removes the constant. Ratios of weighted expectations under the sampled measure reproduce conditional expectations under (6) exactly.

Corollary 2 (Conditional-Haar expectations from weighted draws). *Fix θ satisfying the conditions of Lemma 1 with $\kappa(\theta) > 0$. Let $E_z[\cdot \mid \theta]$ denote the expectation under the restricted construction measure of Proposition 1, the distribution of the retained rotations at θ . For every function h integrable under $\pi_v(\cdot \mid \theta, y, SR)$,*

$$\int h(\theta, Q) \pi_v(dQ \mid \theta, y, SR) = \frac{E_z[v(Q; \theta) h(\theta, Q) \mid \theta]}{E_z[v(Q; \theta) \mid \theta]}. \quad (\text{C.15})$$

Proof of Corollary 2. Write $\tilde{\mu}_z(dQ \mid \theta)$ for the restricted construction measure (C.11). By Proposition 1(b), the conditional of (6) has a density proportional to $v(Q; \theta)$,

$$\pi_v(dQ \mid \theta, y, SR) = c(\theta) v(Q; \theta) \tilde{\mu}_z(dQ \mid \theta),$$

³⁰The weight is the part of the importance weight of ARRW's Algorithm 3 that survives at fixed θ . Their importance weight is the ratio of the target normal-generalized-normal density to the density of their Algorithm 2, a product of two volume elements. The element v_{f_h} converts between the structural and the orthogonal reduced-form parameterizations (their Proposition 1). The element $v_{(g \circ f_h) \mid Z}$ corrects their Algorithm 2 on the zero-restricted set (their Theorem 3). Only the second element arises here. The Metropolis step targets the structural posterior of θ directly, and the sampled and target distributions differ only in the conditional of Q given θ . The volume element $v(Q; \theta)$ of Proposition 1 is their $v_{(g \circ f_h) \mid Z}$ at fixed θ .

with a constant $c(\theta) > 0$. For every h integrable under the conditional,

$$\int h(\theta, Q) \pi_v(dQ \mid \theta, y, SR) = c(\theta) \int h(\theta, Q) v(Q; \theta) \tilde{\mu}_z(dQ \mid \theta) = c(\theta) E_z[v(Q; \theta) h(\theta, Q) \mid \theta].$$

The choice $h = 1$ makes the left side one and pins down the constant, $c(\theta) = 1/E_z[v(Q; \theta) \mid \theta]$.

Substituting the constant into the display gives (C.15). $\square$

The rotation draw. With the target of Definition 1 in place, the rotation draw keeps θ fixed and applies the column-wise ESS update of KZ (Proposition 4) through γ_z . The next proposition establishes its validity on $Q(\theta)$.

Proposition 2 (Column-wise ESS on $Q(\theta)$). *Fix θ satisfying the conditions of Lemma 1 with $\kappa(\theta) > 0$, and let $\pi_X(\cdot \mid \theta, SR)$ denote the conditional distribution of X given θ in (C.8). The column-wise ESS update proposes*

$$x_j(\vartheta) = x_j \cos \vartheta + \tau_j \sin \vartheta, \quad \tau_j \sim \mathcal{N}(0, I_n),$$

where x_j denotes the j -th column of the current $X = [x_1, \dots, x_n]$, and checks admissibility through γ_z . It has the following properties.

- (a) *Almost surely, it produces proposals whose rotations satisfy every zero restriction exactly along the entire ellipse, and the zeros never reject a proposal.*
- (b) *Each column update terminates almost surely.*
- (c) *It leaves $\pi_X(\cdot \mid \theta, SR)$ invariant.*

Proof of Proposition 2. We prove the three parts in turn.

Part (a): the zeros never reject a proposal. Fix a column j , and let $X^*(\vartheta)$ denote the current X with column j replaced by $x_j(\vartheta)$. The sampler evaluates admissibility only through the candidate rotation $\gamma_z(X^*(\vartheta); \theta)$. By Lemma 1(i), every candidate satisfies the zero restrictions exactly wherever the

construction is defined. The construction fails only where some projected column in (C.6) equals the zero vector, an exceptional set of inputs with Gaussian probability zero.

Lemma 1(i) concerns a fresh Gaussian input. The candidates along the ellipse are not fresh Gaussian inputs. A Fubini argument extends the statement to the ellipse. Fix an angle with $\sin \vartheta \neq 0$. Given the current state, the candidate column $x_j(\vartheta) = x_j \cos \vartheta + \tau_j \sin \vartheta$ is a full-support Gaussian draw, and the remaining columns arise from draws with densities. The candidate matrix $X^\star(\vartheta)$ lies in the exceptional set with probability zero. By Fubini's theorem over the uniformly drawn angle, with nonnegative integrands, the exceptional angles form a null set almost surely. The zeros hold along the entire ellipse off this null set, and the zeros never reject a proposal. The slice step retains only angles at which the sign and narrative indicator holds, exactly as in KZ's HARS.

Part (b): almost-sure termination. We argue in three steps.

Step 1: the angle $\vartheta = 0$ is admissible. The bracket of the slice search always contains $\vartheta = 0$. At $\vartheta = 0$ the candidate matrix is the current X . By **Lemma 1**(iii), γ_z maps the current X to the current rotation. The chain retains only admissible states, and the current rotation is admissible.

Step 2: $\vartheta \mapsto \gamma_z(X^\star(\vartheta); \theta)$ is continuous at $\vartheta = 0$ almost surely. Each construction step depends on the stacked matrix $M_j(\theta)$ in (C.5), abbreviated M_j , only through the orthogonal projector P_j onto its null space $N_j(\theta)$. The projector P_j does not depend on the choice of orthonormal basis. Wherever M_j has full row rank,

$$P_j = I_n - M_j'(M_j M_j')^{-1} M_j, \quad (\text{C.16})$$

a continuous function of M_j . The proposition fixes θ under the maintained conditions of **Lemma 1**. Those conditions give M_j full row rank at every orthonormal collection of preceding columns whose columns satisfy their zero restrictions. Along the ellipse, the construction steps $1, \dots, j-1$ do not move, and only the construction steps from j onward vary, by **Lemma 1**(ii). Off the exceptional set of part (a), each construction step is the composition of $M_j \mapsto P_j$ through (C.16) and the normalization in (C.6). Both pieces are continuous at almost every angle. The composition of the n

steps is continuous at $\vartheta = 0$ almost surely.

Step 3: the bracket-shrinking rule terminates. The sign and narrative restrictions are strict inequalities. The admissible set of rotations is open. By Steps 1 and 2, the admissible angular set contains an open neighborhood of $\vartheta = 0$ and is nonempty. The bracket-shrinking rule reaches the admissible angular set in finitely many steps almost surely. Steps 1 and 3 make explicit the nonemptiness assertion in the proof of KZ’s Proposition 4. That assertion suffices in their setting. The QR map γ is continuous at every full-rank X . Step 2 supplies the analogous continuity for γ_z , the only new ingredient.

Part (c): invariance. The target (C.8) is a standard Gaussian density multiplied by a nonnegative likelihood factor, here the admissibility indicator. The Gaussian density factorizes across columns, $\varphi(X) \propto \exp(-\frac{1}{2} \sum_{j=1}^n \|x_j\|^2)$, and the full conditional of a single column is

$$p(x_j \mid x_{-j}, \theta, SR) \propto \mathbf{1}\{SR(\theta, \gamma_z(X; \theta)) > 0\} \exp(-\frac{1}{2} \|x_j\|^2),$$

a Gaussian density times an indicator, as in KZ’s Proposition 4. Elliptical slice sampling is exact for targets of this form (Murray et al., 2010). Part (a) reduces the indicator along every ellipse to a sign-and-narrative indicator. Part (b) supplies the almost-sure termination. The argument of KZ’s Proposition 4 applies. Each column update samples from the admissible angular set and satisfies detailed balance for the full conditional $p(x_j \mid x_{-j}, \theta, SR)$. The systematic scan over the columns composes invariant full-conditional updates. The scan leaves $\pi_X(\cdot \mid \theta, SR)$ invariant. The retained states are serially dependent draws from $\pi_X(\cdot \mid \theta, SR)$. This proves the proposition. $\square$

Lemma 1 (ii) permits the column-wise ESS update on $Q(\theta)$. The zeros hold under every elliptical proposal by construction, and the slice check evaluates only sign and narrative admissibility. The angle-zero proposal is the current admissible state. The admissible angular set is nonempty, and the bracket-shrinking rule terminates (Proposition 2(b)).

The validity of Algorithm 1 step by step. The validity of HARS-Z follows step by step from KZ. The posterior draw in Step 1 is the sampler of BPSS, and its invariant distribution is the posterior target of θ . By Proposition 2, the rotation draw in Step 3 leaves the X -block conditional target (C.8) invariant. The retained rotations of Step 3 are serially dependent but are valid draws from the Q -block conditional target (KZ, Definition 1 and Proposition 4), extended to $Q(\theta)$ by Proposition 2. Their distribution at fixed θ is the induced measure of Proposition 1. The global rotation repair in Step 2b targets the same restricted construction measure through independent proposals. By KZ's Theorem 1, the posterior-draw repair in Step 2a redraws a completed parameter candidate and preserves the sign-restricted marginal target of $z_0 = (A_0, \Lambda)$. The sampler never uses the conditional shortcut for z_+ , which would have led to the incorrect marginal target of z_0 .³¹ Because X does not enter the likelihood, by Corollary 1, none of the rotation rebuilds, rotation draws, or repair steps reweights θ . The retained pairs (θ, X) are draws from the target of Definition 1. Where the zeros load on more than one column, the volume weight of Proposition 1(b) restores the conditional $\pi_v(\cdot \mid \theta, y, SR)$.

C.5 Posterior Inference

Let θ_d denote the d -th retained draw of θ , $Q^{(d,r)} = \gamma_z(X^{(d,r)}; \theta_d)$ its r -th retained rotation, and $v_{d,r} = v(Q^{(d,r)}; \theta_d)$ the volume element at that rotation. The estimator of the posterior mean of a function h of the impulse responses is

$$\widehat{E}[h] = \frac{1}{D} \sum_{d=1}^D \frac{\sum_{r=1}^{N_{\text{ess}}} v_{d,r} h(\theta_d, Q^{(d,r)})}{\sum_{r=1}^{N_{\text{ess}}} v_{d,r}}, \quad (\text{C.17})$$

where D is the number of retained draws of θ . The reported quantiles apply the same weights, normalized within each retained draw. The numerator and the denominator of the inner ratio are

³¹The natural alternative holds $z_0 = (A_0, \Lambda)$ from the stale draw fixed and redraws only $z_+ = A_+$ until the check passes. This targets the wrong distribution. Under the sign-restricted posterior, the marginal of z_0 has the admissibility weight $\omega_0(z_0; s_0, Q)$, the probability that a fresh Gibbs draw of A_+ completes the pair admissibly. Redrawing A_+ until success gives every z_0 a completion. It strips this factor and over-represents structural values whose admissible completions are rare (KZ, Theorem 1).

sample averages of the numerator and the denominator on the right side of (C.15) at θ_d . The outer average runs over the posterior draws of θ , whose marginal the rotation block leaves unchanged by Corollary 1. The estimator is consistent for expectations under (6) as D and N_{ess} grow. Any θ -dependent rescaling of the volume element cancels in the inner ratio. In the single-column case of Proposition 1(a), the weights are constant within each draw and (C.17) reduces to the unweighted average of KZ.

D Estimation Diagnostics

The baseline estimates come from one chain of 10,000 retained draws after 1,000 burn-in iterations, with $N_{\text{ess}} = 5$ rotations kept per sweep. The stability mask keeps 9,615 of the draws. The repair budgets are $N_{\theta} = 300$ and $N_Q = 200$.

Sampler diagnostics. The Metropolis acceptance rate is 48 percent. The chain discards the remaining 52 percent of the proposed transitions, the stale rotations that the joint redraw does not repair within the budgets. The joint structural repair succeeds on 1,050 of 372,322 attempts.

The volume weights stay near uniform, with a within-block weight ESS share of 0.997.

Residual diagnostics. Figure D.1 plots the five equations Figure 4 leaves out. Table D.1 repeats that estimation with the four variance regimes of Table 1 in place of the single regime and standardizes each structural residual by the square root of its estimated regime variance λ_{i,s_t} . Under Gaussian errors these standardized residuals would show excess kurtosis near zero.³² Pooled over the eight equations and the full sample they show 5.0, and the SOMA equation alone shows 24.1.

The two-step procedure of Brunnermeier et al. (2021) fits 5.84 degrees of freedom to these residuals, and the estimation uses that value. The fitted distribution matches the residuals on how often extreme residuals occur. In a sample of 1,648 observations it puts the expected number of residuals beyond six standard deviations at 0.6, and the sample holds one. Beyond five standard deviations it expects 1.5, and the sample holds two.

Table D.2 reports the estimated variance ratios $\omega_{i,s} = \lambda_{i,s}/\lambda_{i,1}$ over the retained draws.

The regime profiles differ across equations. The one-year yield variance stays above its Regime 1 level in all three later regimes. The SOMA variance rises elevenfold in Regime 3 and falls to a tenth of Regime 1 in the other two. Those two equations also carry the widest bands, spanning a factor of two or more in at least one regime. The bands of the other six stay inside a factor of 1.7 everywhere.

³²Kurtosis is the expectation of the fourth power of a standardized random variable and equals three for a Gaussian. Excess kurtosis subtracts this benchmark. Positive values indicate that extreme observations occur more often than under a Gaussian. A Student- t with $\delta > 4$ degrees of freedom has excess kurtosis $6/(\delta - 4)$.

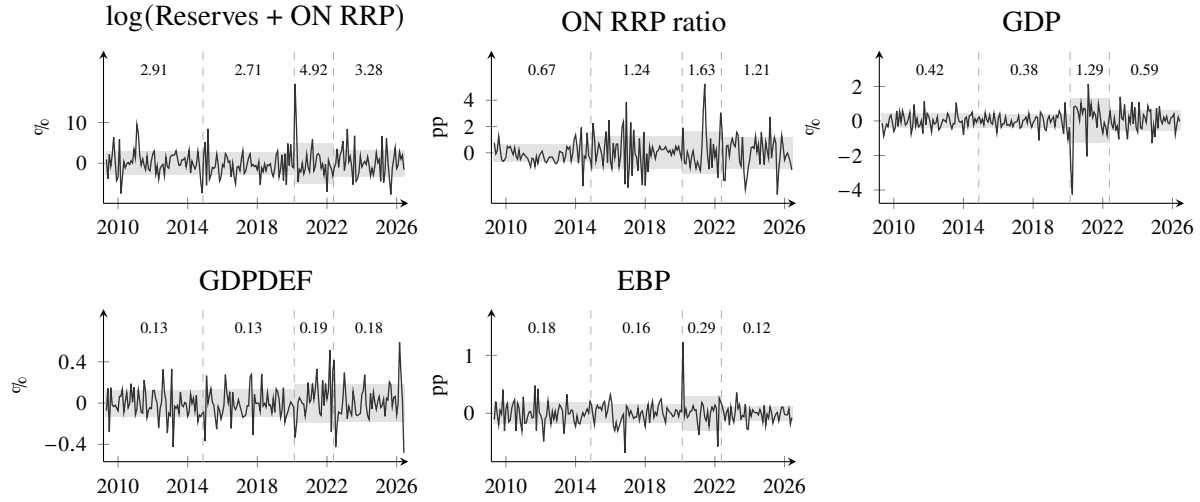


Figure D.1: Reduced-form residuals under a single variance regime, the remaining five equations.
Note. The equations [Figure 4](#) leaves out, on the same estimation and with the same bands and subsample standard deviations. Units are percent for the log levels and percentage points for the ratio and the EBP.

Table D.1: Regime-standardized residuals under four variance regimes.

	Excess kurtosis	$\max z $
One-year yield	3.7	5.5
SOMA	24.1	9.6
log(Reserves + ON RRP)	0.6	3.5
ON RRP ratio	1.1	3.7
GDP	1.5	3.8
GDPDEF	1.3	3.8
Term spread	0.6	3.5
EBP	1.5	4.2
Pooled	5.0	9.6

Note: The table estimates equation (1) with the four variance regimes of [Table 1](#), Gaussian errors, no identification restrictions, and the prior of [Section 3.1](#). It reports the excess kurtosis and the largest absolute value of the structural residuals standardized by $\sqrt{\lambda_{i,s_t}}$. Entries are posterior medians. The pooled row stacks all equations over the full sample.

Table D.2: Estimated variance ratios by regime, $\omega_{i,s} = \lambda_{i,s}/\lambda_{i,1}$.

Equation	Regime 2	Regime 3	Regime 4
One-year yield	11.95 [7.93, 16.26]	65.75 [45.29, 89.55]	49.38 [34.37, 68.13]
SOMA	0.10 [0.06, 0.14]	11.22 [10.60, 12.50]	0.11 [0.07, 0.18]
log(R + RRP)	0.76 [0.75, 0.77]	0.95 [0.94, 0.95]	0.89 [0.86, 0.92]
ON RRP ratio	3.77 [3.46, 4.43]	5.75 [5.23, 6.77]	4.14 [3.76, 4.83]
GDP	0.37 [0.35, 0.40]	1.78 [1.73, 1.82]	0.63 [0.48, 0.80]
GDPDEF	0.95 [0.92, 0.97]	1.81 [1.77, 1.84]	1.95 [1.86, 2.06]
Term spread	0.63 [0.62, 0.64]	0.93 [0.92, 0.93]	1.04 [1.01, 1.06]
EBP	0.78 [0.77, 0.79]	1.38 [1.36, 1.41]	0.41 [0.37, 0.44]

Notes: Entries are posterior medians with 68% credible bands in brackets over the retained draws, weighted by the volume weights. Rows index the structural equations, labeled by the variable on which each equation is normalized. Regime 1 equals one by normalization. Regime dates follow Table 1.

E Impulse Responses to the Demand, Supply, and MP Shocks

Figure E.1 reports the impulse responses to the demand, supply, and MP shocks under the baseline identification.

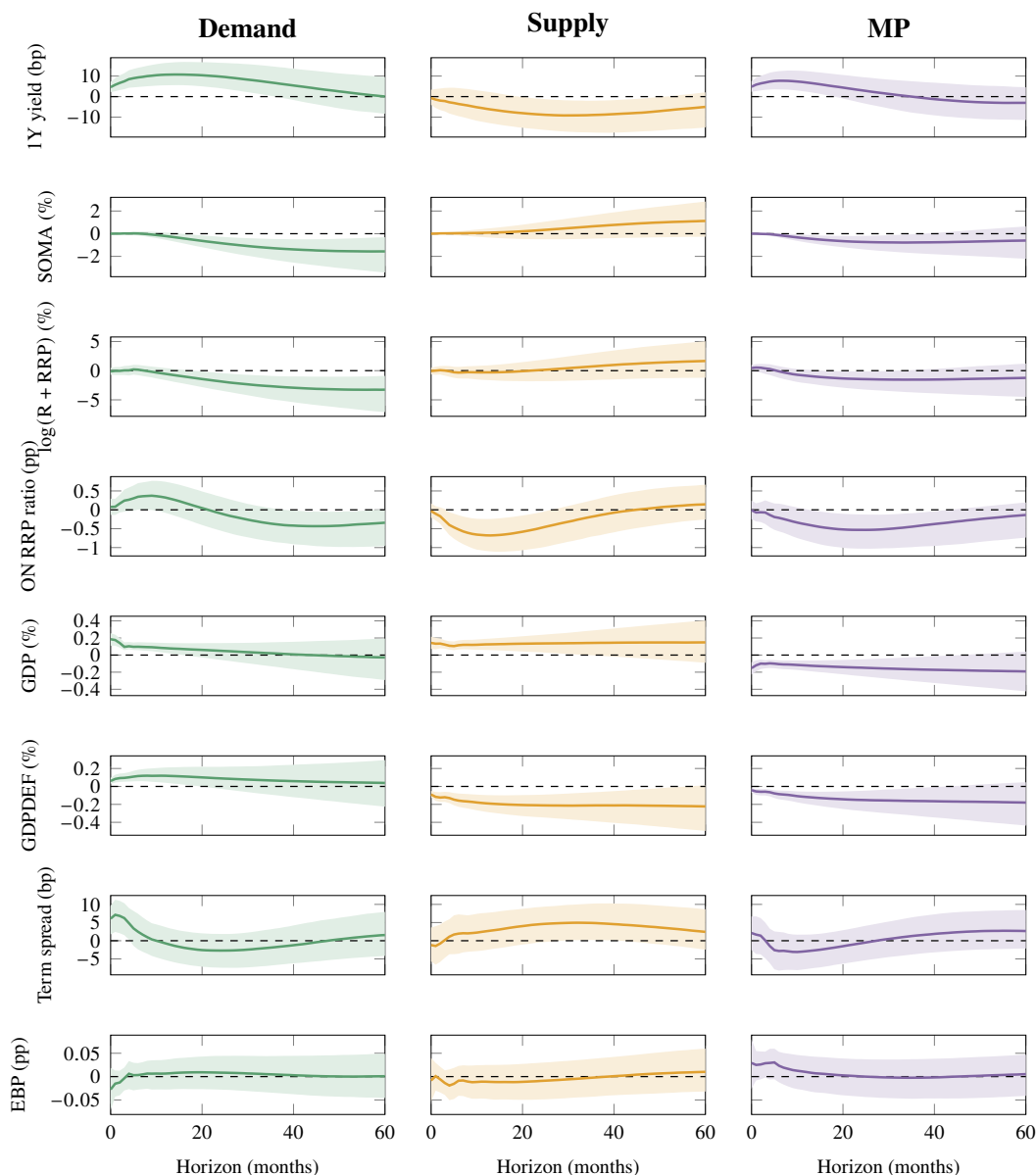


Figure E.1: Impulse responses to the demand, supply, and MP shocks.

Note. Columns report the demand, supply, and MP shocks, and rows report the eight variables with their units. Each shock is scaled to one standard deviation. Under Assumption 2 the SOMA impact response to all three shocks is zero, and the one-percent-SOMA normalization of the balance-sheet figures does not apply. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. Each row shares one vertical range across the three shocks.

F Robustness of the Balance-Sheet Impulse Responses

The figures below plot, for each robustness specification of Section 5.2, the responses of GDP, the deflator, and the EBP to the two shocks that Table 4 summarizes, one figure per specification, in the format of Figures 6 and 7.

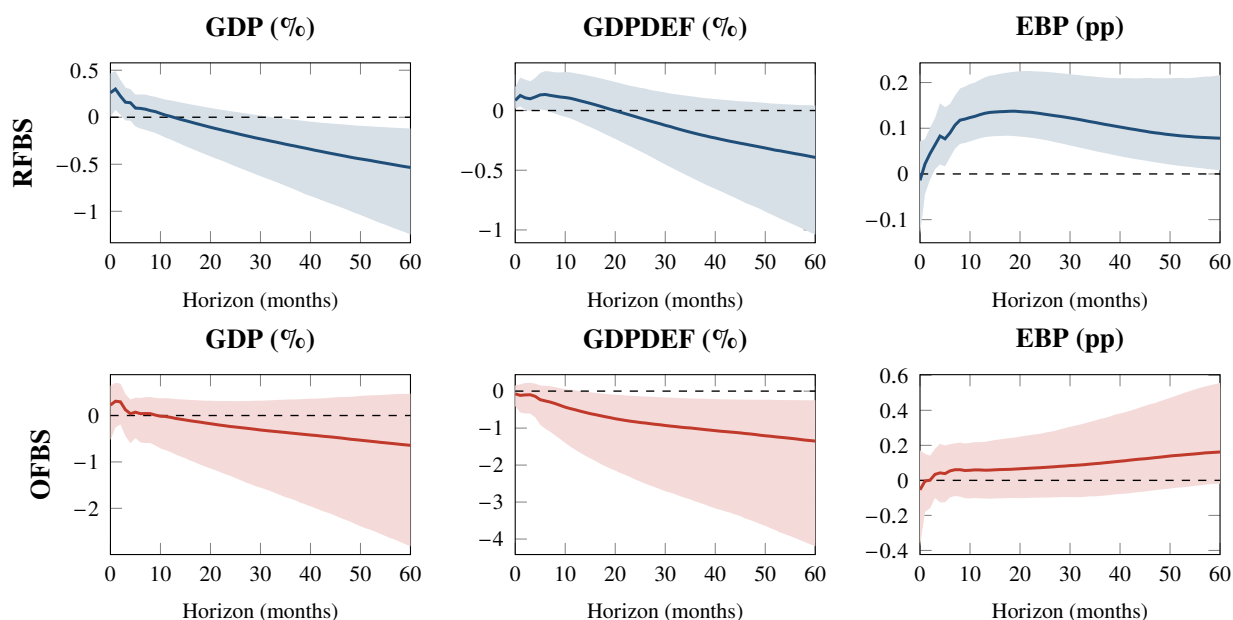


Figure F.1: Robustness with Assumption 1 removed.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The model drops the four impact zeros of Assumption 1 and keeps the three SOMA zeros of Assumption 2. Everything else is as in the baseline.

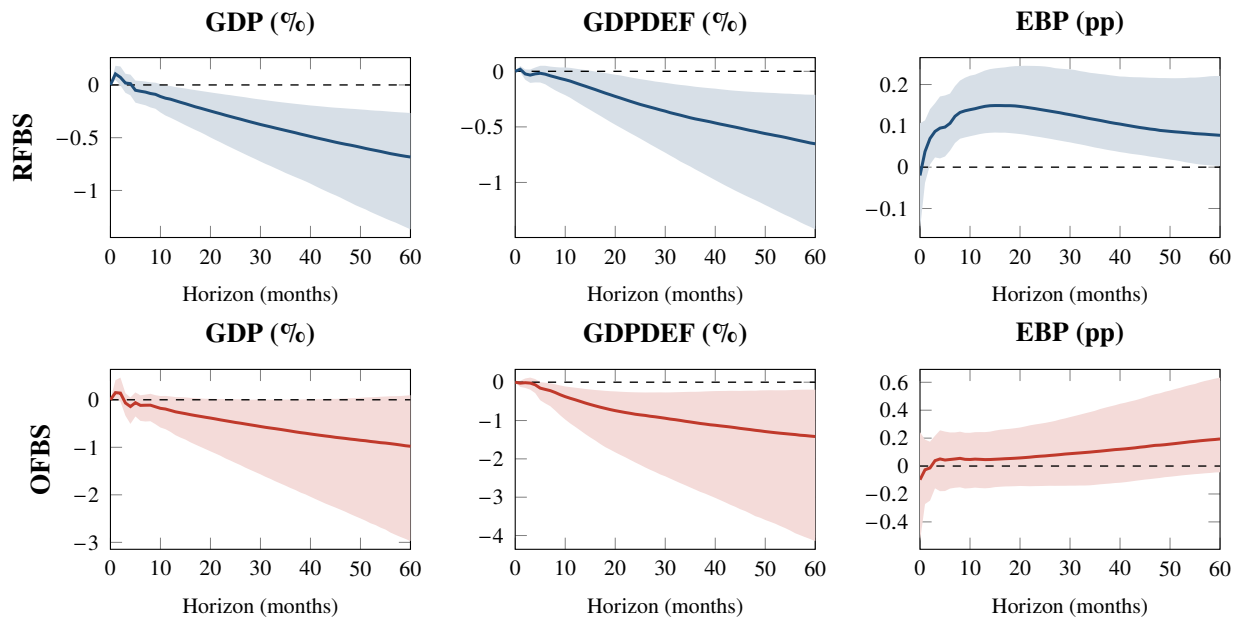


Figure F.2: Robustness with Assumption 2 removed.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The model drops the three SOMA zeros of Assumption 2 and keeps the four impact zeros of Assumption 1. Everything else is as in the baseline.

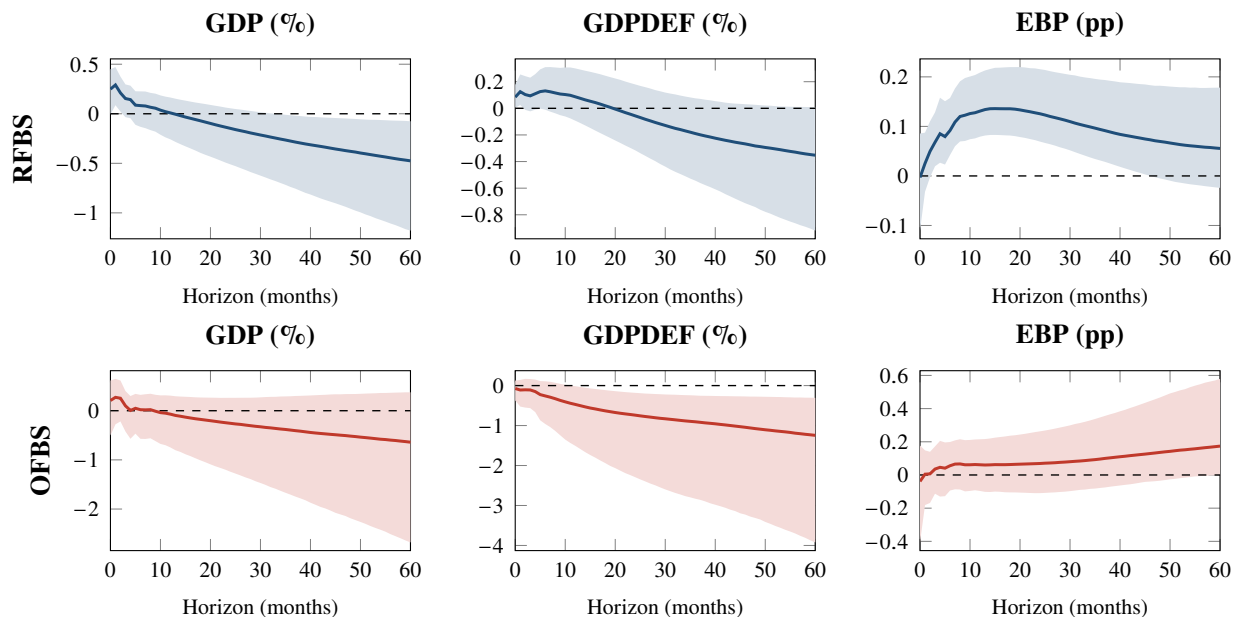


Figure F.3: Robustness with the zero restrictions removed.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The model drops all seven zeros and keeps every sign and narrative restriction. Everything else is as in the baseline.

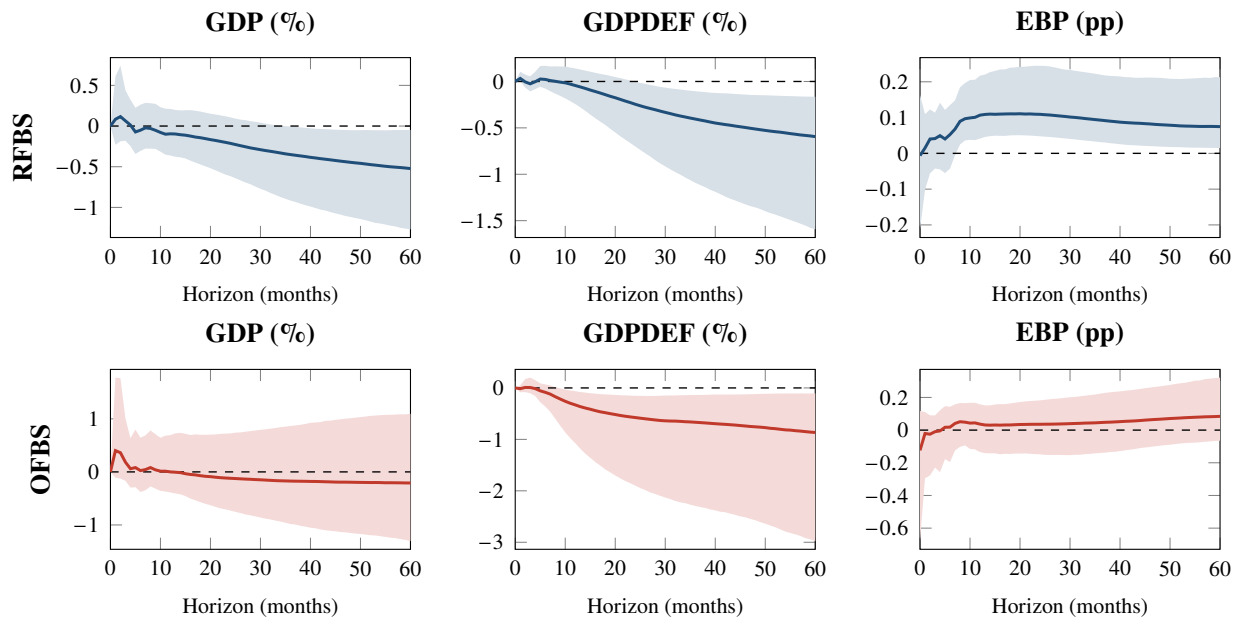


Figure F.4: Robustness with a single variance regime.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The model uses one variance regime and no latent scales, and both narrative anchors fall in that regime. Everything else is as in the baseline.

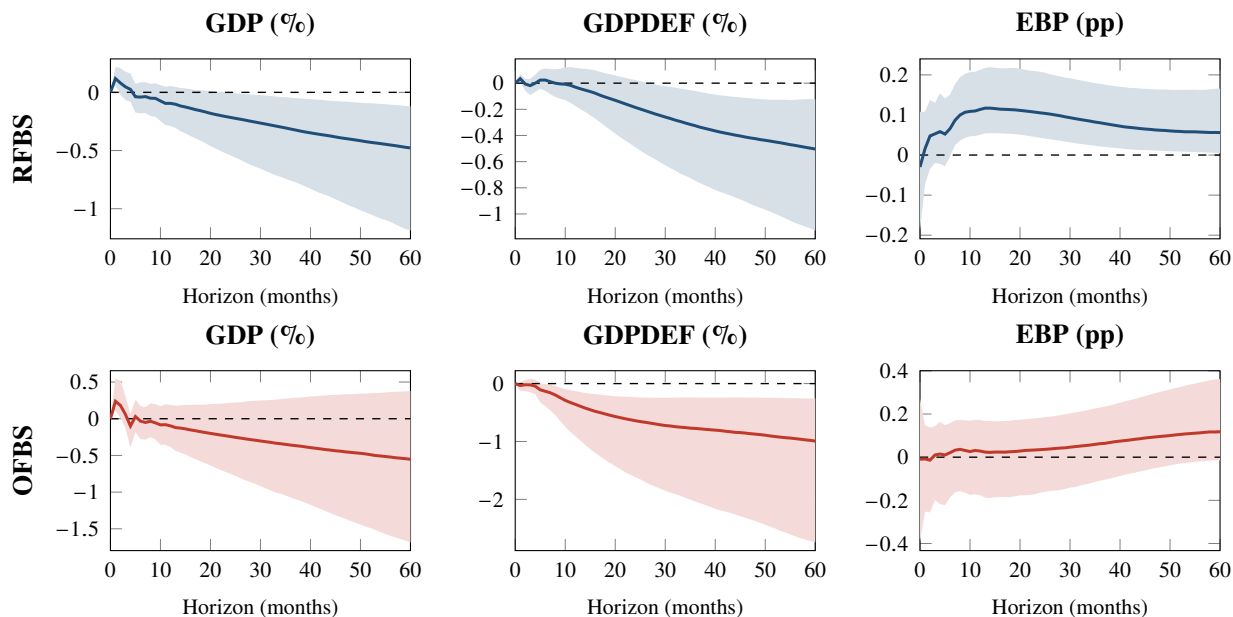


Figure F.5: Robustness with Gaussian errors.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The model removes the latent scales and keeps the four variance regimes. Everything else is as in the baseline.

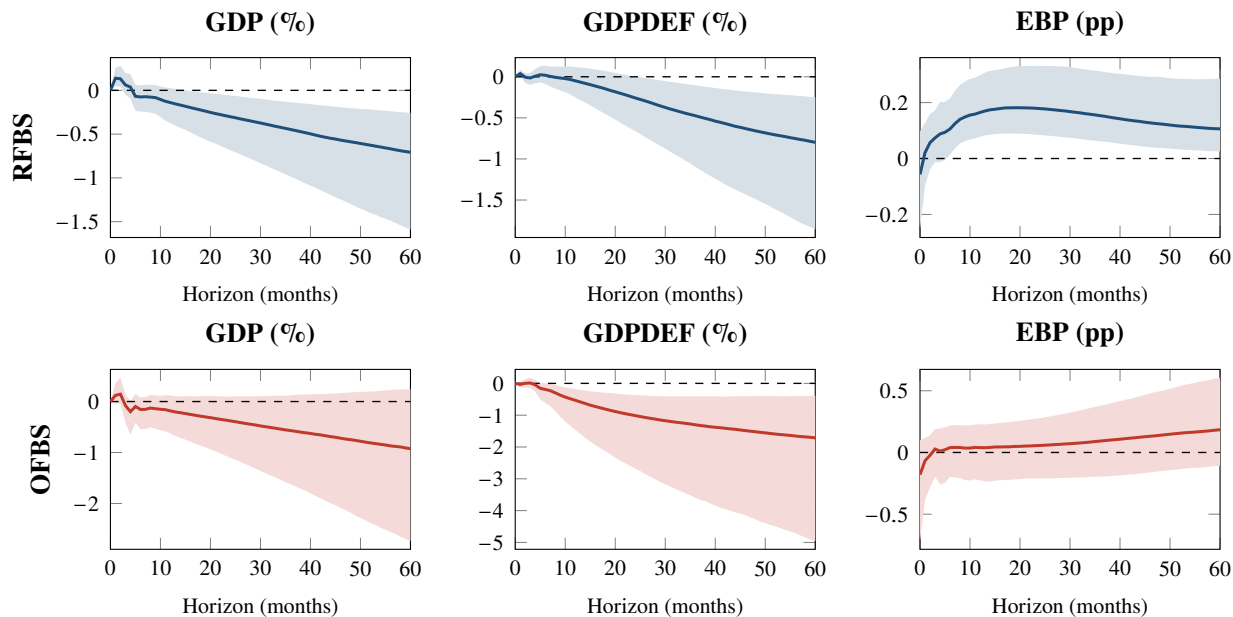


Figure F.6: Robustness with regime breaks at the lower bound.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The regime breaks sit at the exits from and the return to the zero lower bound, 2015:M11, 2020:M3, and 2022:M2. Everything else is as in the baseline.

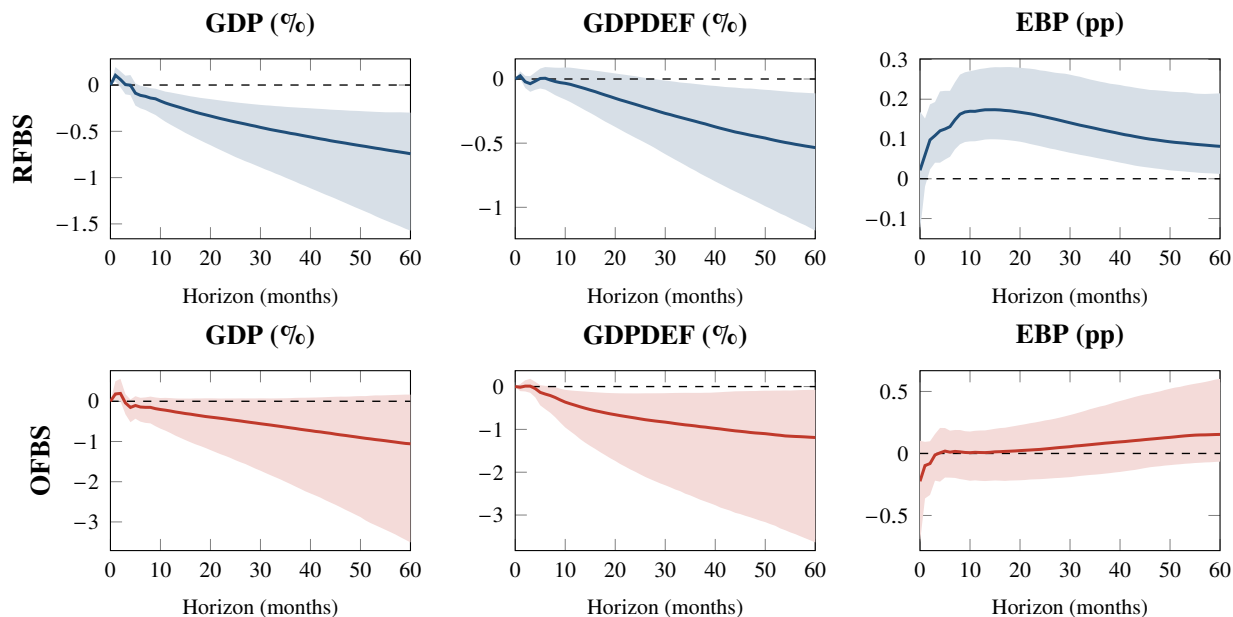


Figure F.7: Robustness with the TGA as an exogenous regressor.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The model adds the ratio of the TGA to the sum of reserves, ON RRP, and the TGA at lags zero to two. Everything else is as in the baseline.

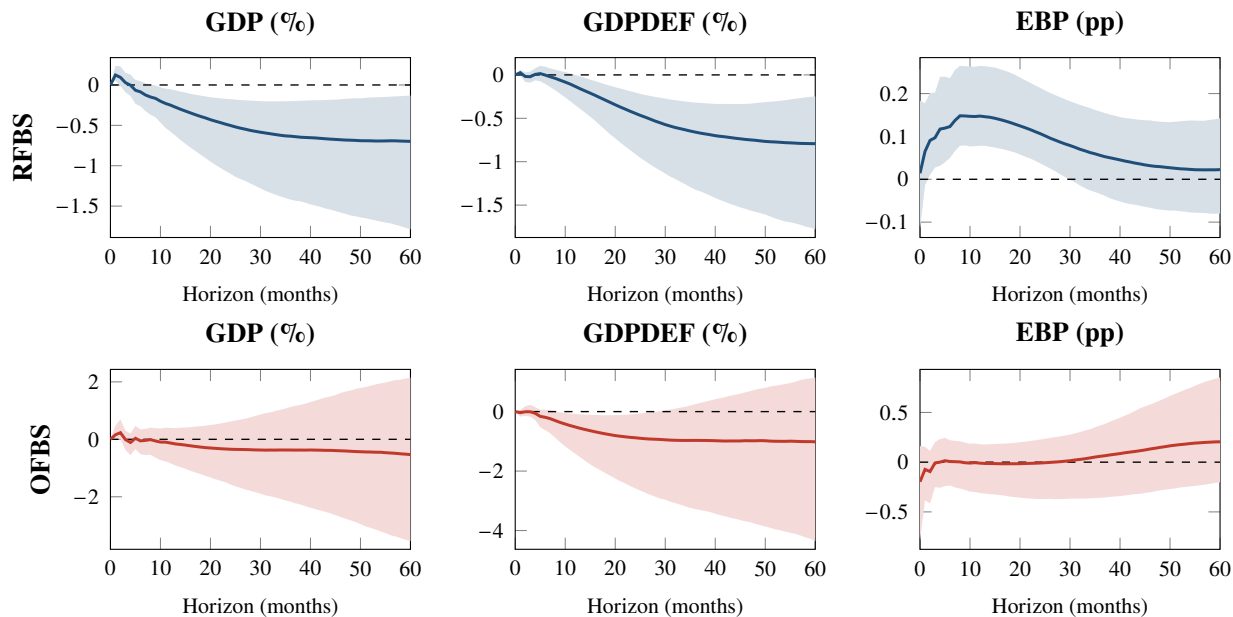


Figure F.8: Robustness with Treasury issuance as exogenous regressors.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The model adds the bill share of marketable debt and the log change of coupon securities outstanding, each at lags zero to two. Everything else is as in the baseline.

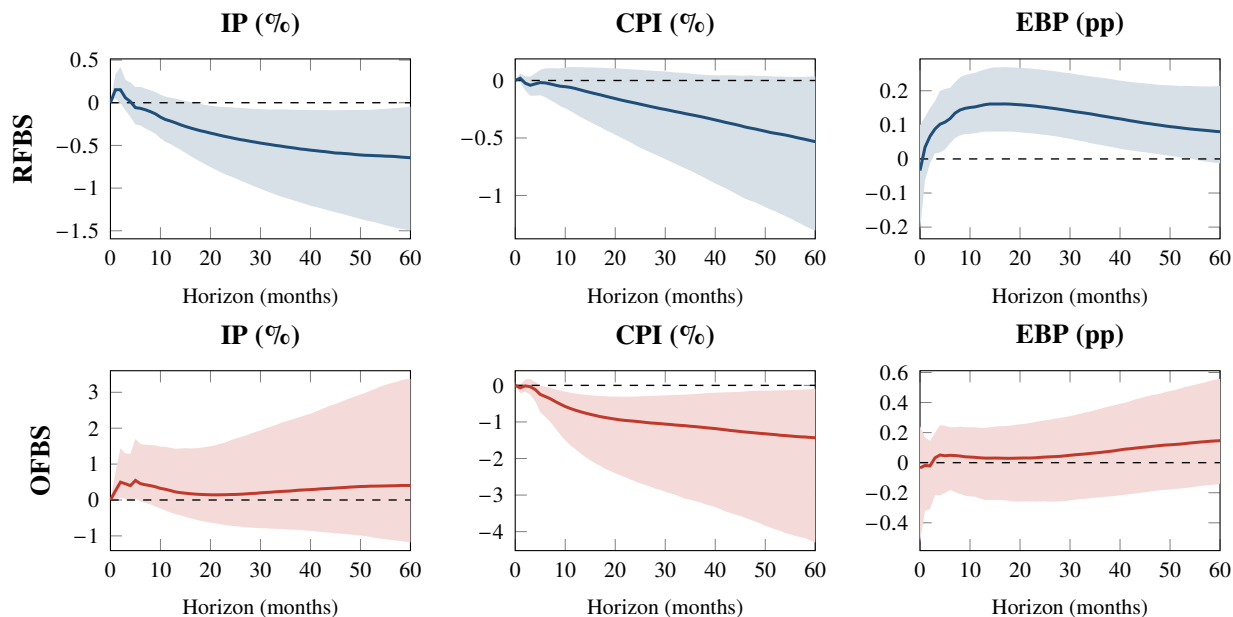


Figure F.9: Robustness with monthly outcome variables.

Note. Rows are the RFBS shock and the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings and shown in the contractionary direction. Solid lines are posterior medians and shaded regions are 68% credible bands over the retained draws. The log of industrial production replaces GDP and the log of the CPI replaces the deflator, and every restriction moves to the new rows. Everything else is as in the baseline.

G Historical Decompositions for All Shocks

Figure G.1 reports the decomposition of Figure 8 for the five variables that figure does not show.

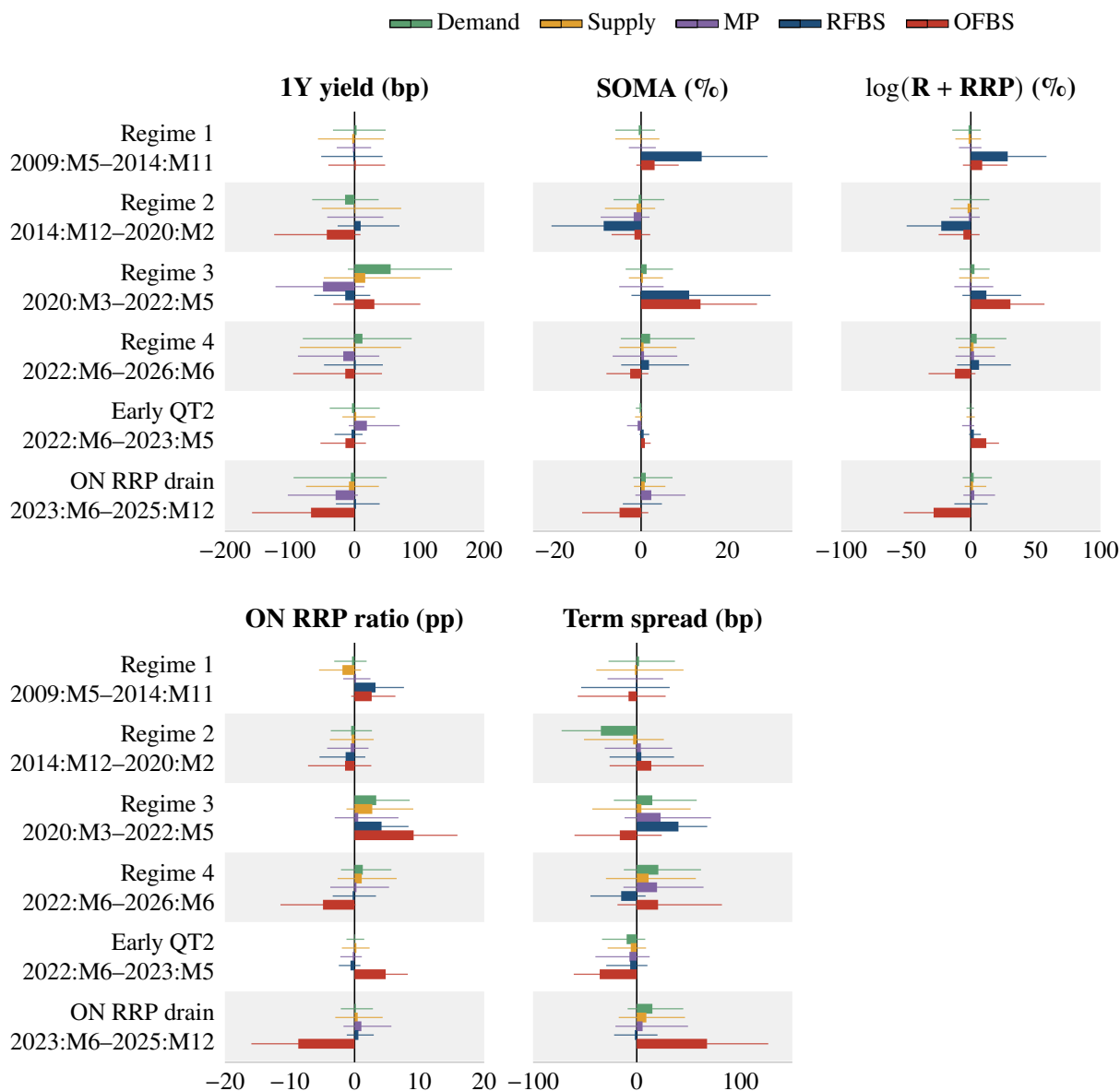


Figure G.1: Historical decomposition by volatility regime, the other five variables.

Note. Contribution of each labeled shock to the accumulated forecast error of the variable over the window. Units are basis points for the one-year yield and the term spread, percent for SOMA holdings and log(Reserves + ON RRP), and percentage points for the ON RRP ratio. Bars are posterior medians and whiskers are 68% credible intervals, both over the retained draws. The windows are those of Figure 8. The three unlabeled shocks account for the remainder. In the size panel the RFBS contribution excludes zero in Regimes 1 and 2 and the OFBS contribution in Regime 3, over early QT2, and over the drain. In the ratio panel the OFBS contribution excludes zero in the same three windows, positive over the buildup and negative over the drain.